

► COMMENT

Colour code for this build. green = comment; red = correction, with deletions struck out in grey; blue = still to be done; purple = suggested new text; orange = the manuscript and the code disagree. Yellow marks every number that will move when the outstanding runs land; yellow + red marks a number currently computed from fewer than twenty realizations, which should not be quoted at all until the runs complete.

Per your status: diesel is treated as final (no highlight), the base case as almost final (light highlight on aggregates only), 2SP as provisional throughout, and the look-ahead sweep as complete except MIPTAIL. All numbers below come from the tables regenerated on 2026-08-07 at 11:25–11:47.

Joint scheduling of driver activities and charging for long distance electric truck operation under travel time uncertainty

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Abstract

Battery electric trucks are entering long-haul freight operations, but various additional constraints compared to conventional diesel trucks discourage freight companies from transitioning their fleets. As charging infrastructure expands, long-haul routes become increasingly feasible, and efficient schedules seek to synchronize charging stops with mandatory driver breaks. However, this synchronization remains fragile due to battery limitations and operational uncertainty, and becoming stranded due to lack of battery is unacceptable. This paper studies the joint scheduling of driver activities, such as breaks and rests, and partial non-linear fast charging activities for a battery electric truck serving a fixed sequence of customers on a long-haul route under travel-time uncertainty.

We formulate a mixed-integer linear program capturing the Hours of Service regulations, including split breaks, reduced rests, extended driving allowances, and the interaction between charging and break. Around this model, we build and compare six decision architectures that differ in when decisions are committed and what information they use: a scenario-based rolling-horizon look-ahead policy, a myopic greedy policy, a robust plan, a budgeted robust plan, a two-stage stochastic plan with online duration recourse, and a perfect-hindsight oracle.

Experiments on synthetic corridors and a real Norwegian long-haul instance show that TO-BE-DONE – policies that adapt schedule en route perform much better than plans that commit at departure – electrification of those corridors cost XXX in route duration compared to diesel truck – decision maker and policies should focus on investment in larger chargers in CS instead of denser network of CS, but that poses problem of grid investment for powerful chargers –

Experiments on 1800 synthetic corridor instances show that policies that adapt the schedule en route perform markedly better than plans committed at departure: the rolling-horizon and two-stage architectures stay within 12 % and 15 % of the hindsight optimum on average, against 25 % for a budgeted robust plan and 35 % for a box robust plan, and the entire penalty of committing early is paid in customer time-window compliance rather than in driving time. Electrifying these corridors costs 8–11 % of route duration against a diesel truck on identical routes and identical travel-time realizations, rising to 20–23 % for an operator scheduling myopically, so roughly half of the observed electrification penalty is a planning failure rather than a physical cost. Finally, decision makers should direct charging-infrastructure investment towards more powerful chargers rather than a denser network: at the spacing already mandated by Regulation (EU) 2023/1804, charger power outweighs charger density by a factor of roughly four, although this shifts the burden onto grid connection capacity at each site.

Keywords: Electric trucks, TDSP, travel time uncertainty, rolling horizon

1. Introduction

Road freight accounts for roughly a quarter of transport CO_2 emissions in the European Union, and battery electric heavy-duty trucks (BEHDTs) are the leading decarbonization pathway for the segment. Their deployment in long-haul service, however, is first constrained by battery capacity and charging infrastructure, but also by an operational factor that is absent for passenger electric vehicles: driver’s regulations. Commercial drivers in the EU operate under Regulation (EC) No 561/2006 and Directive 2002/15/EC, which impose mandatory breaks after at most 4.5 h of driving, daily rest periods, and limits on shift driving time. Because a mandatory 45-minute break is long enough for a substantial fast charge, synchronization of charging and drivers constraints is critical. Whether electric long-haul operation can match diesel route durations therefore hinges on how well charging events can be synchronized with legally required interruptions.

However, this synchronization is fragile. Travel times on long-haul corridors are uncertain, and a delay on a single leg propagates through the entire schedule: it consumes driving-time budget before the planned break location is reached, shifts arrivals outside customer service windows, and, because energy consumption depends on realized speed, perturbs the state of charge with which the truck reaches the next charging opportunity.

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The two risks pull in opposite directions: a *slow* leg threatens Hours-of-Service compliance and the customer window, whereas a *fast* leg is the one that threatens the state of charge, since the consumption rate rises with speed over the whole operating band of a heavy vehicle. A pre-determined schedule that nests charging inside breaks under average travel times may, in execution, force the driver into an unacceptable event such as running out of battery if the energy consumption deviates from anticipated values.

► **COMMENT**

One sentence added, because the opposite-tails structure is the justification for the two-sided budget in Section 6.5 and is currently never stated. See the orange note in Section 2.4 for the numbers.

This paper studies the joint scheduling of driver activities and charging for a battery electric truck on a fixed long-haul route with multiple customer stops, under stochastic travel times. Given the sequence of customers, the decisions are where and for how long to charge (with a nonlinear, piecewise-linearized charging curve and partial charging), where to take breaks and daily rests (including split breaks, reduced rests, and extended driving allowances, together with the rule that a sufficiently long charging stop can count as a break), so as to minimize route duration while respecting the daily provisions of EU HoS regulation, battery limits, and customer service windows.

Because decisions are taken sequentially while uncertainty is revealed leg by leg, the central methodological question is about the decision architecture: when should each decision be committed, and what information should it use? We organize our solution methods along exactly this axis. Three offline methods that commit a plan before departure: a box robust plan over an interval uncertainty set, a budgeted robust plan that protects only against a bounded number of simultaneous worst-case legs, and a two-stage stochastic plan whose activity structure is fixed but whose durations adapt online, and two online architectures that re-decide at every stop: a scenario-based rolling-horizon look-ahead policy and a myopic greedy policy representing typical driver behavior. All methods are evaluated in a common discrete-event simulation, and benchmarked against a perfect-hindsight oracle, so that the value of ~~information and the value of~~ adaptivity can be measured directly.

► **CORRECTION**

“the value of information” is deleted because the VSS/EVPI decomposition that would deliver it has not been run (Section 8). “The value of adaptivity” is delivered, by the base-case comparison. Restore the full phrase only if the VSS harness is run.

Our contributions are as follows. First, we model the full ~~weekly~~**daily** provisions of the EU Hours-of-Service (HoS) regulation in the context of electric trucks, incorporating partial charging along a non-linear charging curve. We formulate this as a MILP and validate it against an independent exact method for the HoS rules, **reproducing the published optimum on 39 of 40 benchmark instances and improving on the 40th**. Second, we analyze the impact of travel-time uncertainty through an extensive set of methods spanning different decision architectures, quantifying the effect of committing to a schedule upfront versus adapting en route as new information is revealed. Third, we quantify the electrification penalty of long-

haul routes by decomposing it into charging time, additional station-access time, and ~~wait time~~the break time that charging discharges. This corroborates the finding of Brockmann and Guajardo [4] that mandatory breaks provide sufficient charging windows most of the time, with charge/break coupling of up to 80%, while showing that the access overheads of a fast-charging stop, which that work does not price, absorb about half of the resulting credit. Finally, we derive policy insights regarding charging-infrastructure investment along long-haul corridors. In particular, that charger power outweighs charger density by a factor of roughly four in its effect on route duration.

► CODE vs. PAPER

“weekly” must become “daily”. MILP.py lines 1241–1250 put weekly accounting explicitly out of scope: no weekly rest, no 56 h weekly driving cap, and the 60 h working week is not enforced (you have already removed that constraint from Section 4 — good). This is the first sentence of the contribution list, so it is the first claim a referee checks, and Garaix et al. [12] already own the weekly TDSP. Daily-only is still a first in this combination.

► COMMENT

Two smaller points. (i) “wait time” does not exist in the decomposition: idle waiting is fixed to zero and Table 15 reads 0.00 in every class. The three real non-charging terms are queueing, manoeuvring and bay repositioning, and the interesting term is the negative one. (ii) “corroborates” undersells you — your own decomposition shows the access overheads eat about half the coupling credit, which is a *qualification* of the closest prior work and worth more to a referee than an agreement with it. The added clause keeps your friendlier framing while claiming it.

► TO DO

“coupling of up to 80%” — the class means are 68–76 % (Table 14) and 80 % is the maximum over time-window cells. Say which. Likewise “a factor of roughly four”: from Section 8.3.2 the ratio is 4.3 against 1.2–1.5, so three to four; “roughly four” is the optimistic end.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 defines the problem and states the modeling assumptions. Section 4 presents the deterministic MILP. Section 6 describes the simulation framework and the six decision architectures. Section 7 details the experimental design, Section 8 reports the computational results, Section 9 discusses their implications. Section 10 concludes.

► COMMENT

The section numbering in the compiled v8 read “Section 6 describes the simulation framework” because the complexity draft occupies Section 5. Once that section is removed (Section 5) the numbering returns to 2–3–4–5–6–7, and this sentence is correct as written.

2. Literature review

2.1. The truck driver scheduling problem

Hours-of-service (HoS) regulations impose constraints on the working and driving schedule of commercial vehicle drivers. In the European Union, Regulation (EC) No 561/2006

[9] specifies maximum consecutive and daily driving times, mandatory break structure and minimum daily rest periods. A similar directive (2002/15/EC) limits total shift working time and restricts work during night hours. Together these rules define the truck driver scheduling problem (TDSP): given a fixed sequence of customer stops, find a feasible and cost-efficient driver schedule complying with all regulatory constraints.

The first work to model government HoS constraints in a scheduling context is Xu et al. [36], who study a rich pickup-and-delivery problem under US rules and conjecture that finding a minimum-cost compliant schedule is NP-hard with multiple time windows. Archetti and Savelsbergh [1] show that the US TDSP with single time windows is solvable in $O(n^3)$ time. For European rules, Drexler and Prescott-Gagnon [6] formalize EU HoS as resource constraints in the elementary shortest path problem with resource constraints (ESPPRC), develop exact and heuristic labeling algorithms, and conjecture NP-completeness. Goel [14] proposes a depth-first label-setting procedure for EU rules, which is extended in Goel [15] to cover extended driving allowances and reduced rest exceptions. Kok et al. [20] present a dynamic programming heuristic for combined vehicle routing and driver scheduling under full EU social legislation, showing that the flexibility of complex break rules yields substantial cost reductions. Prescott-Gagnon et al. [30] embed EU driver rules as resources in a column-generation framework for the VRP with time windows, treating the TDSP as a pricing subproblem. Rancourt et al. [31] address long-haul routing under US HoS with tabu search, confirming that the choice of embedded scheduling algorithm significantly impacts solution quality. Goel and Vidal [17] compare EU, US, Canadian and Australian HoS regulations through optimization, finding that EU rules achieve the highest road safety at a moderate cost premium. Exact branch-and-price algorithms for the combined VRTDSP are developed by Goel and Irnich [16], which is the first with proven optimality guarantees. This method is subsequently accelerated by Tilk and Goel [33] via bidirectional labeling. The most comprehensive EU treatment to date is Garaix et al. [12], who develop a label-setting algorithm for the weekly TDSP and are the first to model the night-work rule, which limits shift working time to ten hours when night work is performed. Across this literature the driver schedule is computed once, against deterministic travel times. To our knowledge no exact EU TDSP formulation has been integrated in a sequential decision framework in which uncertain leg durations are revealed as the route is executed, which is the setting of the present paper.

2.2. The electric vehicle routing problem

The electric vehicle routing problem (EVRP) extends classical vehicle routing to account for the limited driving range of battery electric vehicles and the need for en-route recharging at charging stations. For a comprehensive survey we refer to Erdelić and Carić [7]. The foundational formulation is the green VRP of Erdoğan and Miller-Hooks [8], which routes alternative-fuel vehicles through refueling stations with full recharging. Exact branch-price-and-cut algorithms for the EVRPTW, covering four variants from single full recharges to multiple partial recharges, are developed by Desaulniers et al. [5], who establish that allowing both multiple and partial recharges reduces routing cost and fleet size.

Partial recharging is studied by Felipe et al. [10] for a multi-technology green VRP and it is shown by Keskin and Çatay [18] to substantially reduce route duration. Depot-level charge scheduling over multiple days, accounting for time-dependent energy prices and battery degradation, is studied by Pelletier et al. [26]. The charging function for battery follows a concave nonlinear profile due to the constant-current/constant-voltage physics of lithium-ion cells. While most work simplify this assumption by assuming a linear charging function, it may also be linearized through the use of a piecewise-linear (PWL) approximation [21].

Uncertainty at charging stations has received growing attention: Keskin et al. [19] address the EVRPTW with stochastic waiting times due to limited charger availability, showing that queuing uncertainty significantly affects routing decisions. Energy consumption uncertainty is addressed by Pelletier et al. [27] using a robust optimization framework, demonstrating that robust solutions come at only a modest cost increase. The bulk of EVRP research targets light-duty vehicles in urban or regional settings. Battery electric heavy-duty trucks (BEHDTs) differ qualitatively: much larger energy demands per leg, reliance on high-power fast charging, and binding HoS constraints that govern both when the driver may be on duty and how charging time interacts with mandatory break requirements.

2.3. Combining TDSP and EVRP

Despite the practical importance of long-haul BET operations, the literature combining HoS regulation with battery energy management in a single optimization model is sparse. Bai et al. [2] study the problem of jointly determining optimal charging and rest decisions for an electric truck on a fixed pre-planned route under HoS constraints. They formulate a bilinear MIP minimizing total cost (charging plus rest time) and develop a rollout-based approximate solver that scales linearly with the number of stations, validated on Swedish road network data.

Brockmann and Guajardo [4] develop a mathematical programming model that synchronizes driver break times with truck charging events under EU HoS, validated on real-world data from northwest Germany. Their key finding is that mandatory breaks create natural charging windows sufficient to match diesel route durations; **Section 8.3.3 revisits that finding with an explicit accounting of the station-access overheads it does not price.** Wan et al. [35] study long-haul truck charging planning with time and energy flexibility, proposing a mixed-integer model for opportunity charging at routine stops. Su et al. [32] introduce a stage-expanded graph for minimizing the carbon footprint of long-haul heavy-duty e-truck trips, simultaneously optimizing path, speed, and charging, and show that the optimized charging strategy compounds the emissions benefit of electrification. Vidotte Plaza et al. [34] are the first to jointly optimize charging infrastructure siting and HoS-compliant driver scheduling for long-haul eHDT fleets, though their focus is on strategic network design rather than operational online scheduling. Finally, Niyomphon et al. [24] propose a chance-constrained stochastic MILP for electric freight on fixed long-haul routes with partial recharging and uncertain freight demand, but without HoS constraints.

2.4. Uncertainty in long-haul freight transport

Travel time uncertainty in long-haul road freight creates two risks: a HoS violation when accumulated driving or working time exceeds its limit before a scheduled break, and battery depletion when a leg **is driven faster than nominal and therefore** uses more energy than planned. While stochastic EVRP models have addressed energy consumption uncertainty [27] and queuing uncertainty at chargers [19], travel time uncertainty and its joint effect on HoS compliance and SOC feasibility remains largely unexplored.

► CODE vs. PAPER

The neutral wording you now have is no longer wrong, but it hides the mechanism. Under the ECR curve calibrated in `src/settings.py`, $ECR(v) = A/v + B + Cv^2$ has its *minimum* at $v \approx 61$ km/h and is increasing over the whole realizable band [50, 90] km/h:

ξ	0.889 (fastest)	1.00 (nominal)	1.20	1.60 (slowest)
speed (km/h)	90	80	67	50
energy, 25 km leg (kWh)	29.40	27.49	26.04	26.67

So a *delayed* leg consumes *less* energy than planned. Stranding risk comes from the fast tail, HoS risk from the slow tail. Section 6.5 already relies on this (two independent budgets); Sections 1 and 6.4 did not say it. Note also that the largest possible energy deviation over the whole support is only +7% — which is what makes 2SP’s 15.5% stranding rate a statement about its *plan structure* rather than about energy noise. Worth one sentence in Section 9.

The framework of this type of problem where decisions have to be made sequentially, and adjusted based on realized uncertainty, fits well with the rolling horizon methodology, especially when the multi stage optimization problem would be too complex to solve. The rolling-horizon paradigm is a well-established technique for multi-period optimization under gradual information revelation [13]. The broader literature on dynamic vehicle routing [28] covers stochastic programming, robust optimization, and online policies, and classifies our problem as one with a fully known route structure but sequentially revealed stochastic leg costs.

2.5. Contribution

Table 1 positions the present work against the most closely related papers across ~~eight~~**six** dimensions. This paper is the first to jointly model full EU HoS (including split-break sequencing, shift working-time limits, multiple rest types, and charging-break interactions), partial fast-charging with a nonlinear PWL charging curve, multiple customer stops with soft time windows on a fixed long-haul route, and stochastic travel times.

~~CHECK AGAIN TABLE~~

► TO DO

The table has six comparison columns, not eight (corrected above). More importantly, **it has no column for what the paper actually contributes**. The manuscript claims the decision-architecture comparison as its methodological contribution, but the table has no

“uncertainty handling” and no “benchmark” dimension, so a referee reading only the table concludes the contribution is “EU HoS + non-linear partial charging” — which Brockmann and Guajardo [4] largely has. Two extra columns would fix it (“Uncert. handling”: deterministic / robust / stochastic / online; “Bound”: exact / hindsight / none). Not applied here, since you asked to keep the current structure.

Second: the rolling-horizon and dynamic-VRP strand cited above has no row at all, so the new column would have nothing to contrast against. Consider adding one representative entry.

Table 1: Positioning of the present work in the literature.

Reference	HoS	Charging	Route length	Fixed	TW	Uncertainty
Goel and Irnich 16	EU	—	Long-haul	—	Hard	—
Garaix et al. 12	EU	—	Long-haul	✓	Hard	—
Montoya et al. 21	—	Partial + NL	Regional	—	Hard	—
Keskin et al. 19	—	Partial + L	Long-haul	—	Hard	CS queues
Niyomphon et al. 24	—	Partial + L	Long-haul	—	—	Demand
Bai et al. 2	EU	Partial+L	Day	✓	—	—
Brockmann and Guajardo 4	EU	Partial+NL	Day	✓	—	—
Vidotte Plaza et al. 34	Brazil	Full+L	Long-haul	—	Hard	—
This paper	EU	Partial+NL	Long-haul	✓	Soft	Travel time

Charging: L/NL: linear/non-linear charging function. **TW:** customer time windows.

3. Problem Definition

This paper studies the problem of scheduling driver activities for long-distance freight transport with a battery electric truck (BET). The schedule must simultaneously satisfy Hours-of-Service (HoS) regulations and battery energy constraints. Given a predefined sequence of customer visits, the objective is to minimize the total route duration by determining the optimal timing and locations of mandatory breaks, daily rest periods, and battery charging operations.

► SUGGESTED TEXT

Missing: a short assumptions list. Suggested as a new §3.1, before “Route and Customers”. Everything in it is already true of the code; two items (no idle waiting, deterministic queue) do real work in Section 8.3.3 and are currently discoverable only by reading the source.

3.1 Modelling assumptions. (A1) The order of customer visits is given; routing is out of scope. (A2) One truck executes one route between two weekly rests; fleet effects and depot slow charging are excluded. (A3) There is no idle waiting: service starts on arrival and the only way to delay a departure is to declare a break or a rest, so an early arrival is penalised rather than absorbed. (A4) Only the daily provisions of Reg. 561/2006 and Dir. 2002/15/EC are modelled; weekly rests, the 56 h weekly driving cap and the 60 h working week are out of scope, and $\bar{\rho}$ and $\bar{\epsilon}$ are budgeted per route, which is exact when the route is completed between

two weekly rests. (A5) Energy follows a speed-dependent rate evaluated at the realized leg-average speed; grade, payload variation and the extra consumption of stop-and-go traffic at a given average speed are not represented. (A6) Charger access delay Q_i is a known instance parameter, identical for every method; charger availability and reservation are out of scope. (A7) The objective is makespan plus a fixed service-level penalty; energy price, driver wage and charging tariffs are not priced.

3.1. Route and Customers

The route connects a sequence of customer locations via road segments whose lengths are known in advance. Travel on each segment is characterized by an average speed. In practice, however, traffic conditions introduce uncertainty in the actual travel duration: this is explicitly accounted for in the model. The total route duration is the sum of five components: driving time accumulated while the vehicle is moving between locations, service time for loading or unloading operations, charging time at charging stations, station access and manoeuvring time, and legally required interruptions such as mandatory breaks and daily rest periods.

The sequence of customers to be visited is fixed and known prior to departure. At each customer location, the driver performs a loading or unloading operation whose duration is known. At each customer, soft time windows are considered. Service begins immediately upon arrival, whether or not the window is open: distribution terminals do not turn a truck away, but an arrival outside the announced window, either early or late, disrupts terminal planning and incurs a fixed service-level penalty, identical for early and late arrivals. Any break or rest taken at a customer stop occurs after service. **The vehicle is not permitted to wait idly for a window to open: an early arrival is served early and penalised, or the driver stretches a break (assumption A3).**

► TO DO

Justify β . The paper’s central uncertainty finding is that uncertainty is paid in windows rather than in driving time, and the size of that finding is proportional to β , which Table 5 sets to 1 h with no argument. `settings.py` already carries `BETA_TW_CLASSES = (1, 2, 5)`. Either run that sweep on the reduced set, or say in one sentence why one hour of makespan is the right price for one missed appointment. The duration-only gaps already reported in parentheses in Table 8 are your insurance either way.

3.2. Charging operations

The BET has a battery of fixed capacity (kWh). Energy consumption is assumed to be dependent of the distance traveled and the vehicle speed, yielding a specific consumption rate expressed in kWh/km. At each CS stop, the driver may charge the battery either partially or fully, and multiple charging stops are permitted throughout the route. Charging is not allowed during customer service operations.

Only high-power fast charging is available along the route (overnight slow charging such as in depot is not considered). The charging process is modeled by a piecewise linear approximation of the nonlinear charge-rate curve. Queuing at a CS is modeled as a known

waiting time at each CS, drawn once at instance generation and visible to every method, so it is a deterministic cost of using a station rather than a source of uncertainty.

Finally, a sufficiently long charging stop may simultaneously satisfy a mandatory break requirement. Specifically, a charging event counts as a driving break provided its duration meets the applicable threshold: at least 15 minutes for a first split break, at least 30 minutes for a second split break, or at least 45 minutes for a full break, as defined by the HoS regulations described below.

► TO DO

Give the legal basis for “charging counts as a break” in a footnote: Art. 7 of Reg. 561/2006 requires only that the driver not drive and be free to dispose of the time, and supervising a charge that requires no intervention is generally read as compatible. This is the load-bearing assumption of the whole paper and it is currently asserted without authority.

3.3. HoS Regulations

The scheduling problem is subject to European HoS regulations given by Regulation (EC) No 561/2006 (driving times, breaks and rests) and Directive 2002/15/EC (working time). We model the daily provisions of both instruments; weekly-level accounting is outside the scope of this study, and the reduced-rest and extended-driving allowances are consequently budgeted per route (assumption A4). Two accumulation counters are maintained: driving time, which counts only periods when the vehicle is in motion, and working time, which additionally includes customer service, charger access and maneuvering, and charging time not covered by a simultaneous break.

Regarding mandatory breaks, a driver may not accumulate more than 4.5 hours of continuous driving without taking a full break of at least 45 minutes. This full break may be replaced by a split sequence consisting of a first period of at least 15 minutes (but less than 45 minutes) followed, in that order, by a second period of at least 30 minutes.

Regarding shift-level constraints, the maximum permitted driving time per shift is 9 hours, extendable to 10 hours on no more than two occasions per week. A daily rest of at least 11 h must separate consecutive shifts, and it may be reduced to 9 h at most three times between weekly rests. As a consequence of these rest requirements, the sum of working time, break time, and any other on-duty periods within a 24-hour window is bounded by 13 hours under a full rest, or by 15 hours when the rest is reduced to 9 hours.

Charging during rests is treated as sequential: the truck must vacate the charging bay before the rest begins, and such charging time counts as work.

While charging is only possible at charging stations, breaks and rests may be taken at customer stops, charging stations, and dedicated rest-area nodes inserted along legs.

4. Deterministic Mathematical Model

We now present the mathematical formulation for the deterministic problem. Tables 2 and 3 present all notation used in the model formulation. Figure 2 present a possible solution of the problem.

Table 2: Notation summary.

Sets	
$\mathcal{I}$	Set of possible stops, ordered from 0=origin to N=destination.
$\mathcal{C}$	Set of customers $\mathcal{C} \subseteq \mathcal{I}$.
$\mathcal{K}$	Set of charging stations $\mathcal{K} \subseteq \mathcal{I}$.
$\mathcal{L}$	Set of rest areas $\mathcal{L} \subseteq \mathcal{I}$, $\mathcal{C} \cap \mathcal{K} \cap \mathcal{L} = \emptyset$.
$\mathcal{R}$	Set of breakpoints of the piecewise-linear charging function.
Parameters	
D_i	Travel time from stop i to stop $i + 1$.
E_i	Energy consumption between stop i and stop $i + 1$.
E^{cap}	Battery capacity.
$E^{\min}$	Minimum SOC.
E^0	Initial SOC.
$\bar{E}_r$	SOC level at breakpoint $r \in \mathcal{R}$ of the charging fct, ($\bar{E}_0 = 0$, $\bar{E}_R = E^{cap}$).
$\mathcal{T}_r$	Time to charge from empty to $\bar{E}_r$, ($\mathcal{T}_0 = 0$).
$\mathcal{T}_{\min}$	Minimum charging time at a CS (= 15 min).
S_i	Service time at customer $i \in \mathcal{C}$.
Q_i	Queue time at CS stop $i \in \mathcal{K}$ if charging occurs. Zero if $y_i = 0$.
M_i^{stop}	Maneuver time at CS stop $i \in \mathcal{S}$ if any activity occurs.
M_i^{seq}	Repositioning time at stop $i \in \mathcal{K}$ when charging and a break or rest are declared sequentially, i.e. the truck must vacate the charging bay before the break begins.
W_i^{Ha}, W_i^{Hf}	Service time window at customer $i \in \mathcal{C}$.
β	Fixed out-of-window service penalty (h).
T_{b45}^{brk}	Full break duration (= 45 min).
T_{b15}^{brk}	Min first split break (= 15 min).
T_{b30}^{brk}	Min second split break (= 30 min).
T_{r1}^{rst}	Minimum daily rest duration (= 11 h).
T_{r2}^{rst}	Reduced daily rest duration (exception 1) (= 9 h at most 3×/week).
$\bar{\rho}$	Weekly n Number of allowed reduced rests per route (= 3).
T_{cons}^{drv}	Max consecutive driving time (= 4.5 h).
T_{shift1}^{drv}	Max shift driving time (= 9 h).
T_{shift2}^{drv}	Increased shift driving time (exception 2) (= 10 h at most 2×/week).
$\bar{\epsilon}$	Weekly n Number of allowed extended driving shifts per route (= 2).
$T_{sh1}^{wk} T_1^{spr}$	Max shift working times spread under a regular daily rest (= 13 h).
$T_{sh2}^{wk} T_2^{spr}$	Max shift working times spread under a reduced daily rest (= 15 h).

Abbreviations: SOC = state of charge, CS = charging station

► COMMENT

Three notation fixes. (i) $\mathcal{T}_{\min}$ was listed but never given a value anywhere; MILP.py fixes it at 15 min (`tauc[i] >= 0.25*y[i]`). (ii) $T_{sh1}^{wk}, T_{sh2}^{wk}$ here are the same objects the constraint section calls T_1^{spr}, T_2^{spr} , and they bound the *spread*, not the working time — two names and two meanings for one parameter. (iii) $\bar{\rho}$ and $\bar{\epsilon}$ are described as weekly but enforced once per route in (37) and (52).

Table 3: Variables notation summary.

Variables	
$t_i^a, t_i^d \geq 0$	arrival/departure time at stop $i \in \mathcal{I}$.
$\delta_i \in \{0, 1\}$	1 if service at customer $i \in \mathcal{C}$ starts outside its window.
$y_i \in \{0, 1\}$	1 if the truck charges at stop $i \in \mathcal{K}$.
$\tau_i^c \geq 0$	charging duration at stop $i \in \mathcal{K}$.
$e_i^a, e_i^d \geq 0$	SOC at arrival/departure at CS at stop $i \in \mathcal{I}$.
$\lambda_{i,r}^a, \lambda_{i,r}^d \geq 0$	PWL weight for encoding $e_i^a, e_i^d, i \in \mathcal{K}, r \in \mathcal{R}$.
$\mu_{i,r}^a, \mu_{i,r}^d \in \{0, 1\}$	1 if e_i^a, e_i^d lies in segment r of the PWL curve, $i \in \mathcal{K}, r \in \mathcal{R}$.
$c_i^d \geq 0$	accumulated consecutive driving time at arrival at stop i (h).
$s_i^d \geq 0$	accumulated shift driving time at arrival at stop i (h).
$s_i^w \geq 0$	accumulated shift working time at arrival at stop i (h).
$h_i \geq 0$	Elapsed time since end of last daily rest (h).
$x_i^{b45}, x_i^{b15}, x_i^{b30} \in \{0, 1\}$	break type indicators at stop i .
$\phi_i \in \{0, 1\}$	1 if the first split break has been taken since the last driving reset.
$\tau_i^b \geq 0$	Break time beyond charging at stop i .
$\hat{\tau}_i^b \geq 0$	Break duration credited for HoS purposes.
$g_i \geq 0$	Charging time credited as break at stop i .
$\sigma_i \in \{0, 1\}$	1 if CS i operates in sequential mode (no charging/break synchronization).
$\rho_i^{r1}, \rho_i^{r2} \in \{0, 1\}$	rest type indicators at stop i .
$\tau_i^r \geq 0$	rest duration at stop i .
$v_i \in \{0, 1\}$	1 if any activity (charge/break/rest) occurs at stop i .
$z_i, q_i \in \{0, 1\}$	extended shift flag; extension consumed at a rest.

Abbreviations: HoS = Hours-of-service, CS = Charging station

4.1. Sets and Parameters

Let the route consist of a fixed sequence of stops indexed by $i \in \mathcal{I} = \{0, 1, \dots, N\}$, where stop 0 is the origin and stop N is the destination. The set of customer stops is $\mathcal{C} \subseteq \mathcal{I}$, and the set of CS stops is $\mathcal{K} \subseteq \mathcal{I}$. Additionally, rest areas $\mathcal{L} \subseteq \mathcal{I}$ are place along the route and are treated as Charging station but with charging disabled. The set of non-customer stops $\mathcal{S}$ include both charging station and rest areas, such that $\mathcal{C} \cap \mathcal{K} \cap \mathcal{L} = \emptyset$ and $\mathcal{K} \cup \mathcal{L} = \mathcal{S}$. The model structure is similar to that of Bai et al. [2], adapted here to a single-vehicle fixed-route setting with multiple customer stops, as seen in Figure 1. The nonlinear charging curve is approximated by a piecewise linear function defined by $R+1$ breakpoints, following Montoya et al. [21].

FIGURE NOT IN THE REPOSITORY — `figures/routeExample.png`

Present on Overleaf; reproduced here as a placeholder only.

Your caption is kept below and is correct.

Figure 1: Structure of a benchmark corridor. Charging stations (triangles) are placed on a 60 km grid, rest-area nodes (ticks) subdivide any leg longer than 25 km and allow for a break or rest but no charging, customers (diamonds) are drawn around cluster centers at 25%, 50% and 70% of the corridor. The instance shown is a short-route case: 900 km, three customers.

4.2. Objective Function

The objective minimizes the arrival time at the destination plus a fixed penalty β for every customer whose service starts outside its window.

$$\min \quad t_N^a + \beta \sum_{i \in \mathcal{C}} \delta_i \quad (1)$$

► COMMENT

A referee at TR-E will ask why the objective is makespan rather than cost. The answer is defensible — on a fixed route with a fixed vehicle, driver time dominates the marginal cost and asset utilisation scales with makespan — but it should be said here in one sentence, and the headline number converted once into money in Section 9.

4.3. Constraints

In order to simplify the following constraints, we define additional variables as such: $\rho_i = \rho_i^{r1} + \rho_i^{r2} \in \{0, 1\}$ is the rest indicator at a stop, $x_i = x_i^{b45} + x_i^{b15} + x_i^{b30} \in \{0, 1\}$ is the break indicator at a stop, $r_i = x_i^{b45} + x_i^{b30} + \rho_i^{r1} + \rho_i^{r2} \in \{0, 1\}$ is the reset indicator of consecutive drive time at a stop. We further write $o_i = t_i^d - t_i^a - \tau_i^r$ for the on-duty (non-rest) time spent at stop i , used in the worktime shift counter.

4.3.1. Time propagation

Constraint (2) states that the arrival time at each stop equals the departure time from the previous stop plus the leg travel time. The departure time from a stop depends on the activities undertaken there. At a customer stop, constraint (3) accounts for mandatory service time S_i and any additional break or rest duration. At a CS stop, constraint (4) additionally includes a queue time $Q_i y_i$, a charging duration τ_i^c , a maneuver time M_i^{stop} to get to/leave the CS in case we stop there for any activity, a maneuver time M_i^{seq} if we first charge and then sequentially take a break or rest at the CS. Constraints (5)–(6) activate the out-of-window indicator δ_i whenever service starts before the window opens or after it closes. Since service starts at arrival, the window applies to t_i^a .

$$t_{i+1}^a = t_i^d + D_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (2)$$

$$t_i^d = t_i^a + S_i + \tau_i^b + \tau_i^r, \quad i \in \mathcal{C} \quad (3)$$

$$t_i^d = t_i^a + M_i^{stop} v_i + Q_i y_i + \tau_i^c + M_i^{seq} \sigma_i + \tau_i^b + \tau_i^r, \quad i \in \mathcal{S} \quad (4)$$

$$t_i^a \leq W_i^f + M \delta_i, \quad i \in \mathcal{C} \quad (5)$$

$$t_i^a \geq W_i^a - M \delta_i, \quad i \in \mathcal{C} \quad (6)$$

4.3.2. Battery state-of-charge

Constraint (7) propagates the SOC between stops according to the energy consumed on each leg. Constraints (8)–(9) enforce that charging can only occur at designated CS stops and can never decrease the SOC. Constraints (10)–(11) keep the SOC within the feasible battery operating range at all times. Constraint (12) ensures that a positive charging duration can only be incurred when the binary y_i is active, and that in case of charging, we charge at least as much as the minimum charging time allowed.

$$e_{i+1}^a = e_i^d - E_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (7)$$

$$e_i^d = e_i^a, \quad i \in \mathcal{C} \cup \mathcal{L} \quad (8)$$

$$e_i^d \geq e_i^a, \quad i \in \mathcal{K} \quad (9)$$

$$e_i^a \geq E^{\min}, \quad i \in \mathcal{I} \quad (10)$$

$$e_i^d \leq E^{\text{cap}}, \quad i \in \mathcal{I} \quad (11)$$

$$\mathcal{T}_{\min} y_i \leq \tau_i^c \leq \mathcal{T}_R y_i, \quad i \in \mathcal{K} \quad (12)$$

► COMMENT

$E^{\min} = 0.2 E^{\text{cap}}$ is a modelling choice (range anxiety plus battery health), not a physical limit, and it removes 100 kWh of usable capacity — roughly one 90 km leg. Worth one justifying sentence with a citation in Section 7, since every stranding rate in Table 9 is measured against this threshold rather than against an empty battery.

4.3.3. Piecewise-Linear Charging Function

Following Montoya et al. [21], we linearize the nonlinear charging function using a convex-combination formulation. Constraints (13)–(14) express both the arrival and departure SOC at each CS stop as convex combinations of the $R + 1$ PWL breakpoints. The charging duration is obtained in constraint (15) as the difference in cumulative charge time between the departure and arrival SOC levels. Constraints (16)–(17) impose the convex-combination normalization condition on the arrival and departure weights. Constraints (18)–(19) enforce that exactly one PWL segment is active for each SOC level. Finally, the SOS2 adjacency constraints (20)–(22) restrict non-zero weights to at most two consecutive breakpoints, for both arrival ($\nu = a$) and departure ($\nu = d$) SOC levels.

$$e_i^a = \sum_{r \in \mathcal{R}} \lambda_{i,r}^a \bar{E}_r, \quad i \in \mathcal{K} \quad (13)$$

$$e_i^d = \sum_{r \in \mathcal{R}} \lambda_{i,r}^d \bar{E}_r, \quad i \in \mathcal{K} \quad (14)$$

$$\tau_i^c = \sum_{r \in \mathcal{R}} \lambda_{i,r}^d \mathcal{T}_r - \sum_{r \in \mathcal{R}} \lambda_{i,r}^a \mathcal{T}_r, \quad i \in \mathcal{K} \quad (15)$$

$$\sum_{r \in \mathcal{R}} \lambda_{i,r}^a = 1, \quad i \in \mathcal{K} \quad (16)$$

$$\sum_{r \in \mathcal{R}} \lambda_{i,r}^d = 1, \quad i \in \mathcal{K} \quad (17)$$

$$\sum_{r=1}^R \mu_{i,r}^a = 1, \quad i \in \mathcal{K} \quad (18)$$

$$\sum_{r=1}^R \mu_{i,r}^d = 1, \quad i \in \mathcal{K} \quad (19)$$

$$\lambda_{i,0}^\nu \leq \mu_{i,1}^\nu, \quad (20)$$

$$\lambda_{i,r}^\nu \leq \mu_{i,r}^\nu + \mu_{i,r+1}^\nu, \quad r = 1, \dots, R-1, \quad (21)$$

$$\lambda_{i,R}^\nu \leq \mu_{i,R}^\nu, \quad (22)$$

for all $i \in \mathcal{K}$ and $\nu \in \{a, d\}$.

4.3.4. Break and Rest Decisions

Constraint (23) ensures that at most one type of break or rest activity is declared at each stop.

$$x_i + \rho_i \leq 1, \quad i \in \mathcal{I} \quad (23)$$

At a CS stop, charging time may qualify toward HoS break requirements only when a break is declared and runs in parallel with charging ($x_i = 1, \sigma_i = 0$). Let g_i denote the credited charging time. Then constraints (24)–(27) force $g_i = \tau_i^c$ exactly in that case and $g_i = 0$ otherwise. The effective break duration $\hat{\tau}_i^b$ is then the credited charging time plus any additional break time (28), and the additional break time alone at customer and rest-area stops (29).

$$g_i \leq \tau_i^c \quad i \in \mathcal{K} \quad (24)$$

$$g_i \leq \mathcal{T}_R(1 - \sigma_i) \quad i \in \mathcal{K} \quad (25)$$

$$g_i \leq \mathcal{T}_R x_i \quad i \in \mathcal{K} \quad (26)$$

$$g_i \geq \tau_i^c - \mathcal{T}_R \sigma_i - \mathcal{T}_R(1 - x_i) \quad i \in \mathcal{K} \quad (27)$$

$$\hat{\tau}_i^b = g_i + \tau_i^b, \quad i \in \mathcal{K} \quad (28)$$

$$\hat{\tau}_i^b = \tau_i^b, \quad i \in \mathcal{C} \cup \mathcal{L} \quad (29)$$

► SUGGESTED TEXT

This block is the formulation’s real contribution and it is stated only algebraically. One worked sentence would carry it: “A 50-minute charge at a station where a full break is declared in parallel discharges that break at zero marginal time cost; a 30-minute charge does not, so σ_i is forced to one and the 45-minute break is appended after the charge at a cost of $M_i^{seq} + 45$ min. The whole diesel comparison of Section 8.3.3 is an accounting of this mechanism.”

Constraints (30)–(32) impose the minimum effective break duration for each break type. Similarly, constraints (33)–(34) impose the minimum rest duration for each rest type. Constraints (35)–(36) force the break and rest durations to zero when no corresponding activity is declared. Constraint (37) enforces that the reduced daily rest of 9 h may be used at most $\bar{\rho}$ times ~~per week~~ **per route**.

$$\hat{\tau}_i^b \geq T_{45}^{\text{brk}} x_i^{b_{45}}, \quad i \in \mathcal{I} \quad (30)$$

$$\hat{\tau}_i^b \geq T_{15}^{\text{brk}} x_i^{b_{15}}, \quad i \in \mathcal{I} \quad (31)$$

$$\hat{\tau}_i^b \geq T_{30}^{\text{brk}} x_i^{b_{30}}, \quad i \in \mathcal{I} \quad (32)$$

$$\tau_i^r \geq T_1^{\text{rst}} \rho_i^{r_1}, \quad i \in \mathcal{I} \quad (33)$$

$$\tau_i^r \geq T_2^{\text{rst}} \rho_i^{r_2}, \quad i \in \mathcal{I} \quad (34)$$

$$\tau_i^b \leq M x_i, \quad i \in \mathcal{I} \quad (35)$$

$$\tau_i^r \leq M \rho_i, \quad i \in \mathcal{I} \quad (36)$$

$$\sum_{i \in \mathcal{I}} \rho_i^{r_2} \leq \bar{\rho} \quad (37)$$

EU regulation allows the standard 45 min break to be replaced by a split break taken in two parts: a first part of at least 15 min followed by a second part of at least 30 min, in that strict order. The binary ϕ_i tracks whether the first part of a split break has been completed since the last consecutive-driving reset. Constraint (38) ensures the second part can only be declared if the first part was taken at an earlier stop. Constraints (39)–(41) propagate ϕ between stops. It is set to 1 when a first split-break part is taken and no reset event has occurred, it is cleared to 0 by any consecutive-driving reset (a full break, the completion of the second split-break part, or any rest).

$$x_i^{b_{30}} \leq \phi_i, \quad i \in \mathcal{I} \quad (38)$$

$$\phi_{i+1} \geq \phi_i + x_i^{b_{15}} - r_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (39)$$

$$\phi_{i+1} \leq \phi_i + x_i^{b_{15}}, \quad i \in \mathcal{I} \setminus \{N\} \quad (40)$$

$$\phi_{i+1} \leq 1 - r_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (41)$$

Constraint (42) flags any activity at a stop of $\mathcal{S}$. Sequential mode σ_i is possible only when charging occurs (43). It is forced whenever charging coexists with a rest (44), since the truck must vacate the charging bay before an 11 h or 9 h rest, and it is forced whenever a break is declared that the charge does not fully cover (45), since the driver must then take the break separately after charging. In (45), $\underline{b}_i = T_{45}^{brk} x_i^{b_{45}} + T_{15}^{brk} x_i^{b_{15}} + T_{30}^{brk} x_i^{b_{30}}$ is the required minimum of the break declared at stop i ; if the credited charge g_i (equivalently, since $g_i = \tau_i^c$ under parallel mode, the charge itself) falls short of it, σ_i is forced to 1, and constraint (25) then sets $g_i = 0$ so that the full break sits sequentially after the charge.

$$\frac{1}{2}(y_i + \rho_i + x_i) \leq v_i \leq y_i + \rho_i + x_i, \quad i \in \mathcal{S} \quad (42)$$

$$\sigma_i \leq y_i, \quad i \in \mathcal{K} \quad (43)$$

$$\sigma_i \geq y_i + \rho_i - 1, \quad i \in \mathcal{K} \quad (44)$$

$$\sigma_i \geq \frac{\underline{b}_i - \tau_i^c}{T_{45}^{brk}} - (1 - y_i), \quad i \in \mathcal{K} \quad (45)$$

4.3.5. Driving regulation

The accumulated consecutive driving time c_i^d increases with each driven leg and is reset to zero by any qualifying break or rest (i.e. whenever $r_i = 1$). Defining the auxiliary $l_i^1 = r_i c_i^d$ as the portion of c_i^d that is zeroed out at stop i , constraint (46) expresses the propagation rule. Constraints (A.1)–(A.3) linearize the bilinear product $l_i^1 = r_i c_i^d$ via a standard big- M reformulation. Constraint (47) enforces the regulatory upper bound on consecutive driving time.

$$c_{i+1}^d = c_i^d + D_i - l_i^1, \quad i \in \mathcal{I} \setminus \{N\} \quad (46)$$

$$c_i^d \leq T_{\text{cons}}^{\text{drv}}, \quad i \in \mathcal{I} \quad (47)$$

The accumulated shift driving time s_i^d is reset only by a daily rest (a break alone is insufficient). Defining $l_i^2 = \rho_i s_i^d$, constraint (48) gives the propagation rule. Constraints (A.4)–(A.6) linearize $l_i^2 = \rho_i s_i^d$ with big- M $T_{sh2}^{\text{drv}} = 10$ h, the largest value s_i^d can legitimately attain on an extended shift. Constraint (49) imposes the maximum shift driving time, conditional on the extension flag z_i . Constraint (50) lets the flag persist within a shift and reset only after a rest. Constraints (51)–(52) charge one unit of the weekly budget $\bar{\epsilon}$ per extended shift, the term z_N covering an unfinished final shift.

$$s_{i+1}^d = s_i^d + D_i - l_i^2, \quad i \in \mathcal{I} \setminus \{N\} \quad (48)$$

$$s_i^d \leq T_{sh1}^{\text{drv}} + (T_{sh2}^{\text{drv}} - T_{sh1}^{\text{drv}})z_i, \quad i \in \mathcal{I}. \quad (49)$$

$$z_{i+1} \geq z_i - \rho_i, \quad i \in \mathcal{I} \setminus \{N\} \quad (50)$$

$$q_i \geq z_i + \rho_i - 1, \quad i \in \mathcal{I} \quad (51)$$

$$\sum_{i \in \mathcal{I}} q_i + z_N \leq \bar{\epsilon} \quad (52)$$

4.3.6. Working regulation

Working time comprises driving, customer service, CS queuing, maneuvering, and charging time not covered by a synchronized break. The work contribution of charging is simply $\tau_i^c - g_i$, which is linear, so no auxiliary variable is needed. The accumulated shift working time s_i^w is reset only by a daily rest. By defining $l_i^4 = \rho_i s_i^w$, constraints (A.7)–(A.9) provide the linearization with big-M $T_2^{spr} = 15$ h (within a shift, working time is bounded only by the spread). Constraint (53) propagates s_i^w . The counter carries no daily cap: the daily-level bound on on-duty time is the shift spread (54)–(56) below, and the counter itself is retained for reporting and for the ex-post verification of the Directive’s working-time provisions.

$$s_{i+1}^w = s_i^w - l_i^4 + D_i + \begin{cases} M_{i+1}^{stop} v_{i+1} + Q_{i+1} y_{i+1} + (\tau_{i+1}^c - g_{i+1}) + M_{i+1}^{seq} \sigma_{i+1} & \text{if } i+1 \in \mathcal{S}, \\ S_{i+1} & \text{if } i+1 \in \mathcal{C}, \end{cases} \quad i \in \mathcal{I} \setminus \{N\} \quad (53)$$

► COMMENT

Good — the 60 h weekly working-time constraint has been removed from this version. That was the most dangerous item in the previous draft: it was in the paper but explicitly out of scope in MILP.py, so every reported number had been produced by a different model from the one described. All that remains is to delete T_{60}^{wk} from Table 2 (done above) and the “approach the weekly driving cap” sentence in Section 7.2.

The shift spread h_i measures the elapsed time since the end of the last daily rest. Since a daily rest is always the last activity performed at a stop, the elapsed time from the end of the previous rest to the start of a rest at stop i is exactly $h_i + o_i$, where $o_i = t_i^d - t_i^a - \tau_i^r$ is the on-duty stop time. Constraint (54) propagates the spread, with the reset $l_i^5 = \rho_i(h_i + o_i)$ linearized in (A.10)–(A.12). Constraint (55) bounds the pre-rest spread at $T_1^{spr} = 13$ h when the upcoming rest is regular and $T_2^{spr} = 15$ h when it is reduced, implementing the requirement that the daily rest be completed within 24 h of the end of the previous one. Constraint (56) is the global bound for shift length.

$$h_{i+1} = h_i + o_i + D_i - l_i^5, \quad i \in \mathcal{I} \setminus \{N\} \quad (54)$$

$$h_i + o_i \leq T_1^{spr} + (T_2^{spr} - T_1^{spr}) \rho_i^r + T_2^{spr} (1 - \rho_i), \quad i \in \mathcal{I} \quad (55)$$

$$h_i \leq T_2^{spr}, \quad i \in \mathcal{I} \quad (56)$$

► SUGGESTED TEXT

Promote this to a remark. The equivalence is a genuine, if small, modelling contribution, and it is the one component of the formulation with no counterpart in the reference model of Section 8.1. Suggested:

Remark 1. Enforcing (55) is equivalent to requiring that each daily rest be completed within 24 h of the end of the previous one. A regular rest lasts 11 h and a reduced rest 9 h; since a rest is always the terminal activity at its stop, the 24 h requirement binds the on-duty spread preceding it at $24 - 11 = 13$ h and $24 - 9 = 15$ h respectively. This replaces a rolling-window constraint, which would need time-indexed variables, by a single state variable propagated along the route.

4.3.7. Initial Conditions

Constraints (57)–(59) fix the state at the origin: the truck departs at a given time t^0 , with initial SOC E^0 , and all HoS accumulators and the split-break flag are initialized to zero. No break or rest activity is permitted at the origin or at the destination: $x_0 = \rho_0 = x_N = \rho_N = 0$.

$$t_0^a = t^0, \quad e_0^a = E^0, \quad (57)$$

$$c_0^d = 0, \quad s_0^d = 0, \quad (58)$$

$$s_0^w = 0, \quad \phi_0 = 0. \quad (59)$$

FIGURE NOT IN THE REPOSITORY — [figures/ExampleSolution3.png](#)

Present on Overleaf; reproduced here as a placeholder only.

Figure 2: A feasible schedule on a 950 km corridor with three customers. The four panels share a common time axis. *Route*: the stops used by the solution. *Truck*: driving, parked, and the intervals during which the vehicle occupies a charging bay. *SOC*: state of charge between $E^{\min}$ and E^{cap} . *HoS*: the consecutive-driving and shift-driving accumulators against their limits. *Driver*: the corresponding driver activity. The schedule illustrates the three regimes of (24)–(45): at 284 km a 45-min break runs in parallel with the charge and is discharged at no time cost ($g_i = \tau_i^c$); at 497 km a 9 h reduced rest forces $\sigma_i = 1$, so the truck vacates the bay and the charge earns no credit; at 540 km and 779 km a 15-min and a 30-min charge discharge the two halves of a split break.

► COMMENT

This is the most communicative object in the paper and its caption is one line. A reader who understands it does not need (24)–(45) to follow Section 8.3.3; a reader who does not will struggle throughout. Consider also moving it *before* the model, as a motivating figure.

5. ~~Complexity of the problem (to be removed)~~ [Complexity draft — agreed for removal]

► TO DO

You have marked this section for removal and I agree: TR-E is not a complexity venue, the surrounding NP-hardness results in the literature you cite are themselves conjectures, and a section that visibly ends in “TBD” costs more credibility than the result would buy. The body is omitted from this build to keep the page count down; nothing else in the paper references it.

Salvage one paragraph before deleting. The charging sub-problem argument is worth keeping as a structural remark in Section 4, because it also motivates valid inequality (B.2):

► SUGGESTED TEXT

Remark 2. For a fixed charging pattern y and in the absence of Hours-of-Service constraints, an optimal solution charges at each active station exactly enough to reach the next active station at $E^{\min}$. This follows from the convexity and monotonicity of the cumulative charge-time function $\mathcal{T}(\cdot)$: moving a unit of energy from a later station to an earlier one moves it onto a weakly steeper part of the curve. The optimal charging *pattern* nonetheless remains combinatorial, because each active station carries a fixed access cost $M_i^{\text{stop}} + Q_i$.

6. Solution methods

Introducing uncertainty into a deterministic scheduling model is non-trivial: the optimal schedule depends on travel times and energy consumption that are not yet known at departure. We compare various method to tackle the problem at hand. First a simulation framework is created in Section 6.1 to track the real life progress of a truck, based on some decisions at each stop and realized uncertainty at each leg. The remainder of this section is organized by *when* a method commits to a decision and *what* information it uses to do so, as summarized in Table 4: Section 6.2 and Section 6.3 describe the two adaptive policies, which re-decide at every stop as the trip unfolds, Section 6.4, Section 6.5, and Section 6.6 describe the three anticipative methods, which fix a full schedule before departure. Finally, Section 6.7 introduces the perfect hindsight model that take the realized uncertainty as nominal value and solve the equivalent deterministic MILP as a benchmark for the other methods.

Methodology	Decision setting	Travel time distribution
Rolling-Horizon (LA)	Online	Scenarios
Greedy	Online	Average Nominal
Robust opt. (RO)	Offline	Robust over an interval set
Budgeted robust opt. (ROBU)	Offline	Robust over a budgeted set
2 stage SP	Offline with online recourse	Scenarios
Oracle	Posteriori	Realized values

Table 4: Solution method characteristics

► COMMENT

The greedy rule uses *nominal* values, not an average over a distribution: with the feasibility guard disabled (`GUARD_QUANTILE = None`) it uses no distributional information at all. The re-ordering you already applied, so the table now follows the subsection order, is a real readability gain.

6.1. Simulation Framework

We decouple the planning problem from the execution problem through a simulation framework, illustrated in Figure 3.

The simulation proceeds stop by stop from origin 0 to destination N . At each stop i , the vehicle has arrived with a known state $\Gamma_i = (t_i^a, e_i^a, c_i^d, s_i^d, s_i^w, h_i, \phi_i, n_i^{r2}, n_i^\epsilon)$ that summarizes accumulated time, energy, and HoS counters, where n_i^{r2} and n_i^ϵ count the reduced rests and driving extensions already consumed. A decision method maps this state to an action a_i specifying whether to charge, which break type to take, and which rest type to take. The vehicle executes a_i , departs, and the travel time on leg $i \rightarrow i+1$ is revealed as a random draw, from which the realized energy consumption follows via the ECR curve. The truck state is updated accordingly and the cycle repeats. Early arrival at a customer triggers immediate service, with the out-of-window penalty recorded; **the vehicle cannot wait for a window to open.**

FIGURE NOT IN THE REPOSITORY — [figures/simulation.png](#)

Present on Overleaf; reproduced here as a placeholder only.

Figure 3: The simulation framework. At each stop the realized state Γ_i is passed to a decision selector, which returns an action a_i (charge / break type / rest type). The action is executed, the vehicle departs, and only then is the travel-time multiplier ξ_i of the next leg drawn; the realized energy follows from the ECR curve at the implied speed. All six methods share this loop and this state representation, they differ only in the box labeled “decision selector”.

► COMMENT

Encoding note: in the source this caption contains mojibake — the multiplier renders as ξ_i and the quotation marks as “”. The file has been saved once as UTF-8 and re-read as Latin-1. Same problem in the author names (“C^A^line”, “Universit^A^”). Re-save the .tex as UTF-8 in Overleaf.

The decisions methods are described below and they share the same simulation loop and vehicle state representation, only the decision step differs. The travel-time realizations are recorded throughout and fed to the oracle at the end of each run.

Because uncertainty is revealed only after departure, a leg may retroactively cause a violation that no stop-level action can repair: the consecutive-driving counter may exceed 4.5 h mid-leg or the SOC may fall below acceptable level. The simulator records these events and runs with any violation are marked infeasible.

► SUGGESTED TEXT

This paragraph defines the paper’s most important measurement convention in three lines, and Section 8.2 leans on it heavily. Two sentences to add:

“Two consequences follow. First, a non-zero violation rate is unavoidable for any online policy: a leg whose realized duration exceeds the remaining consecutive-driving budget breaches Article 7 whatever was decided at the preceding stop. Second, an infeasible run has no meaningful route duration and is excluded from every gap statistic, so a method that fails often is scored

only on the realizations it survives; efficiency and reliability must therefore be read together, Table 8 against Table 9.”

6.2. Rolling-Horizon with look-Ahead policy

We adopt a *rolling-horizon* policy in which a new decision is made at each stop, using only the information available at that point and considering a limited horizon to model the future. Let $i \in \{0, \dots, N-1\}$ denote the current stop and let $\Gamma_i = (t_i^a, e_i^a, c_i^d, s_i^d, s_i^w, h_i, \phi_i, n_i^{r2}, n_i^e)$ be the vehicle state at arrival. At each stop the policy executes the following steps, which are summarized in Figure 4.

All structurally feasible actions $a \in \mathcal{A}_i$ are enumerated. An action specifies the charge binary $y \in \{0, 1\}$ (relevant at CS stops only), a break type $b \in \{\emptyset, b_{45}, b_{15}, b_{30}\}$, and a rest type $r \in \{\emptyset, r_1, r_2\}$. Actions that violate structural feasibility (such as a split-break second part b_{30} without a prior b_{15} , or a reduced rest r_2 when the budget is exhausted) are excluded. No probabilistic feasibility guard is applied: an action that would strand the vehicle is not filtered a priori but is scored as infeasible in the scenarios that expose it, and is penalised through $f(a)$. This keeps the policy’s information set identical to the greedy policy’s.

► COMMENT

The pruning sentence you commented out was right to go — `GUARD_QUANTILE = None` disables that guard entirely. But the reader is then left wondering how LA avoids stranding at all, which the purple sentence answers.

Then, an horizon end-stop $h > i$ is determined by advancing along the route for a nominal duration of T_{hor} hours. This end-stop is defined via a simplistic algorithm that defines, based on rest rule, which stop can be attained in the given horizon.

For each travel leg between our current stop i and the horizon end-stop h , a set of S travel-time scenarios $\{\xi^{(k)}\}_{k=1}^S$ is drawn independently for legs $[s, h)$. Each multiplier $\xi_i^{(k)}$ is sampled from the appropriate travel time random distribution. The corresponding scenario energy $E_i^{(k)}$ is derived from the ECR curve evaluated at the implied speed.

Afterward, for each candidate action $a \in \mathcal{A}_i$ and each scenario k , the sub-problem MILP over the horizon $[s, h]$ is solved with a fixed at the current stop and with travel times $D_i^{(k)} = D_i \xi_i^{(k)}$. The subproblem minimizes arrival time at h , subject to the same SOC, HoS, break, rest, and charging constraints as the full-route model. The score of action a is the aggregation of the S sub-problem objectives:

$$f(a) = \frac{1}{S} \sum_{k=1}^S t_h^{a,(k)}(a),$$

with infeasible scenarios penalized by a large constant. ~~Alternative aggregations (worst-case, best-case, CVAR) to be considered~~

► COMMENT

The alternative aggregations are *already implemented* (`criterion = "mean" | "worst" | "best" | "cvar"` in `Simulation.py`, CVaR at level 0.8), so this is an un-run experiment

rather than an open question. Either report it as a one-line robustness check alongside TB0 and MIPTAIL in Section 8.3.1, or delete the sentence.

Finally, the action $a^* = \operatorname{argmin}_{a \in \mathcal{A}_s} f(a)$ is selected, with a tie-breaking rule that favors charging ($y = 1$). A final MILP re-solve on $[s, h]$ with a^* fixed and nominal travel times produces the exact activity durations $(\tau_i^c, \tau_i^b, \tau_i^r)$ executed by the vehicle. The actual travel time to next stop is then drawn from the simulation framework and the vehicle state σ_{i+1} is updated accordingly.

► **COMMENT**

σ is already the sequential-mode binary of (43). Use Γ_{i+1} here, as in Section 6.1. Same slip in Section 6.6 (“the realized state σ_i ”).

FIGURE NOT IN THE REPOSITORY — [figures/RH.png](#)

Present on Overleaf; reproduced here as a placeholder only.

Figure 4: Rolling horizon with look ahead policy.

In order to speed-up the process, only a partially relaxed LP version of the subproblems are solved. The out-of-window indicators δ_i and the next stops decisions remains as integer or binary, as well as all binary decisions at customer stops and the sequential charging binary at all CS. However, to tighten the relaxation, various valid inequalities are added to the formulation of the LP subproblems. They are summarized in [Appendix B](#).

► **COMMENT**

This now matches `MILP.py` exactly. Worth adding *why*, in half a sentence, because it is a real implementation finding: the big-M relaxation of δ_i is so weak that a two-hour window miss over a hundred-hour horizon gives $\delta \approx 0.02$, so a fully relaxed subproblem treats windows as free and the action ranking is silently biased toward window-ignoring actions.

6.3. Greedy heuristic

To provide a baseline representative of current practice, we implement a *greedy* policy that mimics the behavior of a driver operating without planning tools: stop only when forced to, charge to full whenever stopping, and avoid queuing stations known to be busy. Unlike the rolling-horizon policy, the greedy algorithm uses no look-ahead and solves no optimization sub-problem: it evaluates a fixed priority rule at each stop and returns an action immediately. A break can be inserted in the charging time if the duration is long enough.

► **SUGGESTED TEXT**

The rule should be stated so the baseline is reproducible without reading the code: “The rule is evaluated in order at each stop. (1) If the shift driving limit is about to bind, take a daily

rest, reduced if the budget allows. (2) Otherwise, if the consecutive-driving limit is about to bind, take a full break, or the second half of a split break if the first has been taken. (3) Otherwise, if the energy required to reach the next charging station exceeds the usable state of charge, charge. (4) Otherwise, charge to full at any station whose expected access delay is below a threshold. Whenever the truck is stopped at a charging station and the charge is at least as long as the required break, the break is declared in parallel at no additional time cost.” Add one clause noting that Q_i is a shared model parameter, so rule (4) is not privileged information — otherwise a referee will suspect the baseline was handicapped.

6.4. Robust optimization

The deterministic MILP presented above uses nominal travel times D_i on each leg i . In practice, travel times are uncertain and we model this via

$$\tilde{D}_i = D_i \xi_i, \quad \xi_i \in [\underline{\xi}, \bar{\xi}], \quad i \in \mathcal{I}$$

where $\underline{\xi}$ and $\bar{\xi}$ are fixed, physically motivated bounds on the travel-time multiplier rather than a symmetric perturbation: $\bar{\xi} = V^{\text{nom}}/\bar{v}$ from a leg-average congestion floor $\bar{v}$, and $\underline{\xi} = V^{\text{nom}}/\hat{v}$ from the vehicle speed limiter $\hat{v}$ (90 km/h for HGVs), so the support is asymmetric around 1. This is the most conservative member of the budgeted family of Bertsimas and Sim [3], recovered at full budget $\Gamma = N$. We retain the full box deliberately: this robust plan serves as the maximum-commitment plan. It commits every activity and every duration before departure using only the support of the uncertainty, whereas the two-stage program of Section 6.6 exploits distributional information and online duration recourse. Section 6.5 relaxes this box to a budgeted uncertainty set that removes most of this conservatism while retaining a deterministic feasibility guarantee.

Because energy consumption depends on speed through the ECR curve, each $\tilde{D}_i$ also induces an uncertain energy $\tilde{E}_i(\tilde{D}_i)$. We handle this by substituting the worst-case energy $E_i^{\text{max}} = L_i \cdot \text{ECR}(L_i/(D_i \underline{\xi}))$ into the SOC constraints, leaving them otherwise unchanged.

The robust counterpart is then obtained directly by parameter substitution:

$$D_i^{\text{wc}} = D_i \bar{\xi}, \quad E_i^{\text{max}} = L_i \cdot \text{ECR}\left(\frac{L_i}{D_i \underline{\xi}}\right), \quad i = 0, \dots, N-1.$$

The two parameters use opposite extremes of the uncertainty interval: D_i^{wc} uses the slowest realization, which is unfavorable for elapsed time and the Hours-of-Service accumulators, while E_i^{max} uses the fastest realization, which is unfavorable for state of charge since the ECR curve is increasing ~~and convex~~ in speed **over the operating band** $[\bar{v}, \hat{v}] = [50, 90]$ km/h. No single realization attains both corners, so the box counterpart is conservative not only because it deviates every leg simultaneously, but because it deviates each leg in two mutually exclusive directions at once.

► CODE vs. PAPER

$\text{ECR}(v) = A/v + B + Cv^2$ is not globally increasing: with the calibrated coefficients it has a minimum near 61 km/h and decreases below that. It is increasing over $[50, 90]$, which is all the argument needs — but the unqualified claim is false and a referee checking the functional form will catch it.

The resulting feasible set therefore provides a guaranteed feasibility under any realization $\xi_i \in [\underline{\xi}, \bar{\xi}]$. The optimal solution at each stop is saved and used in the simulation framework to evaluate it.

6.5. Budgeted robust optimization

The box counterpart of Section 6.4 prices every leg at its worst case simultaneously. Under independent leg multipliers this joint event has very low probability. We therefore also field the budgeted uncertainty set of Bertsimas and Sim [3],

$$\mathcal{U}_\Gamma = \left\{ \xi : \xi_i = 1 + z_i^+(\bar{\xi} - 1) - z_i^-(1 - \underline{\xi}), \quad z_i^+, z_i^- \in [0, 1]^N, \quad \sum_i z_i^+ \leq \Gamma, \quad \sum_i z_i^- \leq \Gamma \right\},$$

in which at most Γ legs' worth of deviation mass is slow ($\xi_i \rightarrow \bar{\xi}$, as in Section 6.4, unfavorable for elapsed time and the Hours-of-Service accumulators) and, independently, at most Γ legs' worth is fast ($\xi_i \rightarrow \underline{\xi}$, unfavorable for state of charge through the ECR curve). We use the classical protection level $\Gamma = \lceil \sqrt{N} \rceil$ [3]. We use the classical protection level $\Gamma = \lceil 1 + z_{1-\varepsilon} \sqrt{N} \rceil$ with $\varepsilon = 0.01$, for which any single constraint over N independent, symmetrically distributed multipliers is violated with probability at most ε [3]; this gives $\Gamma = 18, 24$ and 31 on the short, medium and long route classes respectively. The budget interpolates between the nominal model ($\Gamma = 0$) and the box of Section 6.4 ($\Gamma = N$).

► CODE vs. PAPER

`RObudget.py:classic_gamma` implements $\lceil 1 + z_{1-\varepsilon} \sqrt{N} \rceil$ with $\varepsilon = 0.01$, not $\sqrt{N}$. The difference is large: on a short route with $N \approx 49$ legs, $\sqrt{N} \approx 7$ whereas the implemented budget is 18, i.e. 37% of all legs. Your source already carries the correct sentence, commented out one line above — swap them. The realized Γ values are now stated, because nothing in the paper currently lets a reader size these models, and because they let the reader check the 3.3% ROBU stranding rate against the 1% the budget nominally buys (Section 9).

Unlike the box, the budgeted counterpart does not reduce to parameter substitution. The model is a state-space formulation whose accumulators (driving since the last break, shift driving, state of charge) reset at stops selected by the binary plan, so each robust constraint involves the worst deviation over a decision-dependent window of legs. Classical constraint-wise dualization then yields products of dual variables and plan binaries and is impractical at our instance sizes [29]. The vehicle routing literature meets the same obstacle inside branch-and-price labeling algorithms [22]. We instead solve the counterpart exactly by scenario generation [23, 37]: a Master problem optimizes the plan against a finite subset $\hat{\mathcal{U}} \subset \mathcal{U}_\Gamma$, the extensive-form model of Section 6.6 with a min-max objective and a single non-anticipative duration vector, each scenario carrying its own copy of the state variables, and a Pessimization step searches $\mathcal{U}_\Gamma$ for a realization that breaks the current plan. Because the stop sequence is fixed, once the plan is fixed every accumulator window is known, and the worst attack on each constraint family is available in closed form: fully deviate the Γ largest legs inside that window (a vertex of $\mathcal{U}_\Gamma$ maximizing a nonnegative linear function). Candidate attacks are generated per continuous-driving window, per shift, per customer-window prefix, and per between-charges segment, and each is certified against the full model

with the plan fixed. Violated scenarios are appended to $\widehat{\mathcal{U}}$ and the master is re-solved. The procedure terminates finitely [37], in practice within a handful of iterations, and the master may be solved to a relative gap without losing convergence [25]. At termination the plan is provably feasible for every realization in $\mathcal{U}_\Gamma$.

As for the box counterpart, the resulting plan commits every activity and every duration before departure. The optimal solution at each stop is saved and executed as is in the simulation framework to evaluate it. By the calibration of Γ , realizations outside the set may occur and render the plan infeasible in the simulation framework.

6.6. Two-stage stochastic program

We extend the deterministic MILP into a two-stage stochastic program (2-SP) with fixed recourse. First-stage decisions are the binary decisions at each stop (charge yes/no, break yes/no, rest yes/no), which are fixed once, before any uncertainty is revealed, and are therefore identical across all scenarios. Second-stage decisions are the continuous variables at each stop (charging time, break time, and rest time), which are free to adapt to each scenario’s realized travel times.

Let $k = 1, \dots, S$ index the scenarios, each a random realization of travel time on every leg, drawn independently and with no correlation assumed between legs (the same sampling assumption used for the rolling-horizon sub-problem in Section 6.2). Denoting the first-stage binaries y, b, r and the second-stage variables $\tau^{c,(k)}, \tau^{b,(k)}, \tau^{r,(k)}$, the model solves

$$\min_{y, b, r \in \{0,1\}} \frac{1}{S} \sum_{k=1}^S (t_N^{a,(k)}(y, b, r) + \beta \sum_{i \in \mathcal{C}} \delta_i^{(k)})$$

subject to the SoC, HoS, break, rest, and charging constraints of the deterministic model holding for every scenario k , with nominal travel times and energies replaced by their scenario-specific values. The objective is therefore the expected arrival time across scenarios.

For evaluation, only the first-stage binary decisions are retained from the offline solve. During simulation, at each stop i the remaining binaries are held fixed and the continuous activity durations are re-optimized by solving the linear program over $[i, N]$ with the realized state σ_i and nominal remaining travel times. Then, only the durations at stop i are executed. If this LP is infeasible, a repair step solves a restricted MILP in which binary activities may be added but not removed (for example if we need an extra CS stop), with the objective to first minimize the number of added activities and then arrival time. If the repair also fails, the run is recorded as a violation of the stochastic plan. The plan’s structure is thus committed offline, its durations adapt online, and its repair frequency is itself a reported metric.

► COMMENT

The repair step is what makes 2SP competitive, and it is also what makes its classification as “offline” questionable: it fires on **99.2%** of executions (Table 9). Saying so here rather than only in the results changes how the reader interprets 2SP’s low short-route gaps from the moment they first see them.

6.7. Perfect-Hindsight Oracle

Unlike the methods above, the oracle is not a candidate policy: it is an ex-post performance bound computed once the simulation run is finished, using information no policy could have had while driving. To assess how much is lost by the lack of advance knowledge, we compare the methodologies against a perfect-hindsight oracle: a lower bound on achievable arrival time given the travel times that actually occurred during the simulation run.

After the simulation has completed, let $\tilde{D}_0, \tilde{D}_1, \dots, \tilde{D}_{N-1}$ be the sequence of travel times that were realized on each leg. The oracle solves the full-route MILP of Section 4 with these values substituted for the nominal times D_i . Because the realized travel times are fixed and known when the problem is solved, this is a deterministic MILP identical in structure to the nominal model. Its optimal value $t_N^{a,*}$ is the earliest possible arrival at the destination that any policy could have achieved, had it known all travel times in advance.

The oracle is not a realizable policy: no algorithm can know $\tilde{D}_i$ before driving leg i . It therefore provides a lower bound against which the optimality gap of any online policy can be measured. The hindsight gap Δ is the relative difference of the full route duration. A small Δ indicates that look-ahead scheduling recovers most of the benefit of perfect foresight. In order to speed up the MILP solve, the rolling horizon solution is used as a first feasible solution in the perfect hindsight model.

► SUGGESTED TEXT

A caveat belongs here, not only in Section 8.2, because it qualifies every number in the paper: “The oracle is itself a mixed-integer program and is not always solved to proven optimality within the two-hour budget. Where it is not, $t_N^{a,*}$ is the best incumbent found, so the reported gaps are distances to that incumbent. On the long-route class the residual dual gap is structural rather than a matter of solver budget — it is driven by the packing of the fifth mandatory rest — so long-route gaps should be read as distances to a schedule certified to within roughly 9%, and never compared across route classes.”

7. Experimental design

This section presents the setting of our experimental setup, in order to provide reproducibility of our results. The full data collection is made available in Github [XXX](#). [\[TODO: public repository URL plus an archived DOI \(Zenodo\); TR-E increasingly expects a citable archive rather than a bare link.\]](#)

7.1. Nominal parameters

All experiments share the vehicle and operational parameters listed in Table 5.

► CODE vs. PAPER

Charging station spacing. `instances.py` places stations on a deterministic 60 km grid and jitters each position by $U[-19, +19]$ km, so realized spacings run roughly $[22, 98]$ km with mean 60. $U[40, 80]$ was never what the generator did, and it contradicts Section 7.2’s own “regular average intervals of 60 km”.

Table 5: Nominal vehicle and operational parameters.

Category	Parameter	
Vehicle	Battery capacity E^{cap}	$0.2 \cdot E^{\text{cap}}$
	Minimum SoC E^{min}	
	Full-charge time (0%–100%) $\mathcal{T}_R$	
	Consumption at nominal speed $\text{ECR}(80)$	
Charging	PWL charging segments	0%, 40%
	PWL breakpoints (SoC)	
	Minimum charging time $\mathcal{T}_{\text{min}}$	
Operations	Nominal average speed	
	Customer service time	
	Maneuver time when seq. operations at CS stop	
	Maneuver time at CS and rest areas stop	
	Out of window penalty β	
	Idle waiting	
Infrastructure	Charging station spacing	$\mathcal{U}[40, 80] \text{ km}$ 60 km grid, jittered $\mathcal{U}[$
	Charging station or rest areas spacing	
	Queue delay at CS (log-normal)	$\mu = 10 \text{ m}$

Charging station spacing and charger power output is taken from the Regulation (EU) 2023/1804, which mandates heavy-duty vehicles charging station every 60 km on the TEN-T core network by 2030. The energy consumption rate (ECR) as a function of speed v (km/h) is modeled from Gao et al. [11] and calibrated with verified consumption values from constructors.

► CODE vs. PAPER

Which ECR reference? The paper cites Gao et al. [11]; `src/settings.py` cites Younes et al. (2020), 10.1016/j.est.2020.101758, for the functional form $\text{ECR}(v) = A/v + B + Cv^2$, cross-checked against Nykvist & Olsson (2021) and NACFE Run-on-Less. These cannot both be the source of the coefficients. [TO DO: resolve, and give A , B , C explicitly — the entire energy dimension rests on three numbers that appear nowhere in the paper.]

Travel time on each leg is uncertain through a multiplicative multiplier $\tilde{D}_i = D_i \xi_i$. The multiplier follows a shifted-lognormal law $\xi_i = \min(\underline{\xi} + Y_i, \bar{\xi})$, where Y_i is lognormal with a base coefficient of variation of 0.15, and the fixed bounds are $\underline{\xi} = V^{\text{nom}}/90 \approx 0.889$ (speed limits) and $\bar{\xi} = V^{\text{nom}}/50 = 1.6$ (a leg-average congestion floor). These same bounds $[\underline{\xi}, \bar{\xi}]$ define the box and budgeted uncertainty sets used by the robust methods of Sections 6.4–6.5. The realized energy on each leg follows from the ECR curve at the implied speed. The location parameter of Y_i is calibrated numerically so that $\mathbb{E}[\xi_i] = 1$ exactly after truncation, which keeps the nominal plan an unbiased benchmark; at the base coefficient of variation the upper cap absorbs about 1 % of the mass. Multipliers are drawn independently across

legs.

► TO DO

Independence across legs is the assumption most exposed to attack. Legs are 25 km and a real congestion event spans several consecutive ones. `settings.py` already implements an AR(1) driver (`TRAVEL_TIME_AR1_RHO`, currently 0). Either run $\rho = 0.4$ on the reduced set — correlation should hurt exactly the methods that assume independence, namely ROBU’s budget calibration and 2SP’s scenario set, and leave LA comparatively unharmed, which would itself be a result — or state the limitation and predict the direction in Section 9. Silence is the one option a referee will not accept.

7.2. Benchmark instances

A set of benchmark instances of increasing complexity is generated along a single long-haul corridor. Charging station stops are placed at regular average intervals of 60 km, customer stops are randomly place around clusters centers inserted along the corridor. The instances are parameterized based on route distance, customer count, and time windows width as summarized in Table 6. For each instance, service windows are generated from the deterministic MILP solved with average travel time: each window is centered on the service start of the found optimal solution, with half-width drawn independently for each customer from the class range in Table 6, **and the centre jittered by $\pm 10\%$ of that half-width**. Instances whose deterministic model is infeasible under nominal data are discarded and regenerated. Each Route/Customers/Time Windows class contains 50 independent and randomly generated instances. The number of stops ~~ranges from 51 on average for the short route instances, to 168 for the long route instances~~ **averages 50 on the short routes, 96 on the medium and 167 on the long ones, of which 16, 32 and 58 are charging stations and 26, 56 and 101 are rest-area nodes.**

► COMMENT

Counts recomputed over all 1 800 generated instance files. Rest-area nodes are the *majority* of stops (52–60 %): they carry no service, no charging and no window, but they carry the full HoS state chain, and they are the main driver of model size. That is currently invisible in the text, and it is also what makes the Γ values of Section 6.5 as large as they are.

The $\pm 10\%$ centre jitter matters too: without it every window would be centred on a schedule the deterministic model can hit exactly, and a referee would suspect the design favours the offline methods.

Each route instance class targets a distinct operational regime: ~~Small~~**Short** instances complete with a single rest, Medium instances span two to three shifts, activating the rest, reduced-rest, and extension mechanisms. ~~Large instances approach the weekly driving cap.~~**Long instances span four to five duty cycles, at which point the weekly provisions outside the scope of this model would begin to bind.**

► COMMENT

Two things: Small/Medium/Large here versus Short/Medium/Long everywhere else, and a reference to a weekly cap the model does not enforce. Note also that “Medium” is simultaneously a route class, a customer class and a window class, so “Medium/Medium/Medium” is a

Table 6: Experimental design. Full factorial, $3 \times 3 \times 4 \times 50 = 1\,800$ instances.

Factor	Levels	Values
Route class	Short	800–1 200 km
	Medium	1 500–2 500 km
	Long	3 000–4 000 km
Customers	Few	1–3
	Medium	4–6
	Many	7–15
Time window (half-width per customer)	None	no window
	Tight	$U[0.5, 1.0]$ h
	Medium	$U[1.0, 3.0]$ h
	Large	$U[3.0, 6.0]$ h
Derived	Stops $ \mathcal{I} $	50 / 96 / 167 (short / medium / long)
	Budget Γ	18 / 24 / 31

real cell in Table 8. Renaming the customer classes C1–3 / C4–6 / C7–15 would remove the collision.

7.3. Norwegian case study

TBD In addition to the synthetic corridors, we evaluate all methods on a real long-haul instance along XXX (XXX km), with XXX customer terminals at XXX cities. Charger locations and rated powers are taken from already accessible charging stations for heavy duty vehicles, while leg distances and nominal speeds are obtained through Google maps API. Travel-time uncertainty on each leg is calibrated from XXX (Statens vegvesen traffic data?)

► TO DO

Decision required now, not at submission. `src/instance_gen/norway_instance.py` is a loader skeleton: the code path is complete but none of the four data files it needs exists (`chargers.csv` from NOBIL, `customers.csv`, `legs.csv` from OSRM, `dispersion.json` from Statens vegvesen). No Norwegian instance has been generated or solved.

Build it: export HDV-capable sites from NOBIL for the E6 corridor, pick 3–5 terminal cities, pull leg distances from OSRM, calibrate the coefficient of variation — and ideally the AR(1) coefficient — from `trafikdata.atlas.vegvesen.no`. That last item would simultaneously discharge the independence objection above, which makes this worth more than a case study normally is. The loader’s own comment warns that corridor gradients make the constant-ECR assumption optimistic here.

Or cut it, and remove “and a real Norwegian long-haul instance” from the abstract. 1 800 synthetic instances on a full factorial is a sufficient computational study; an empty section is not. Either way the abstract must stop promising it — the version proposed on page 1 no longer does.

8. Computational results

The experiments reported in this section have been solved using Gurobi 12.0.2 on a computer equipped with an Intel(R) Xeon(R) Gold 6244 CPU @ 3.60GHz processor and 384 GB of RAM.

► TO DO

Add threads per solve, whether runs were executed concurrently, and the Gurobi licence class. This matters more than usual here, because Section 8.3.1 reports *per-stop* decision time rather than wall clock precisely on the grounds that wall clock across configurations would measure contention. If that is true there, it is true in Table 10 too.

8.1. Validation of the deterministic model on TDSP benchmark instances

Before turning to the stochastic experiments, we verify that the HoS core of the deterministic MILP of Section 4 reproduces known optimal solutions of the pure truck driver scheduling problem. To this end we use the 40 R15-PGLT benchmark instances made available by Garaix et al. [12]², which comprise fixed customer sequences with 2–14 customers, per-stop service times (including a loading operation at the origin depot), hard time windows, and all durations rounded to multiples of 15 minutes. We compare against the published optima of their *no-night* configuration, which enforces exactly the daily provisions of Regulation (EC) No 561/2006 and Directive 2002/15/EC considered in this paper, without the night-work rule.

Since these are diesel instances, the benchmark exercises the HoS machinery of our model in isolation: we disable the battery ($E_i = 0, \mathcal{K} = \emptyset$) and reduce all maneuver, queue, and out-of-window penalty terms to zero, with time windows enforced as hard constraints ($\delta_i = 0$). Furthermore, the reference model allows breaks and rests to preempt driving and service at arbitrary points. We emulate this on the 15-minute by inserting rest-area nodes ($\mathcal{L}$) every 15 minutes of driving and by splitting each service operation into 15-minute segments.

► CORRECTION

Restore the four bridging differences you commented out. This is the paper’s only external validation, and its credibility is exactly the credibility of this list — a referee who cannot see how the two formulations were reconciled cannot take 39/40 at face value. Suggested, condensed to keep the length you have now:

Four further differences are bridged as follows. A zero-length rest-area node before each customer permits the arrive–rest–serve pattern present in several reference optima. Their schedules must terminate with a daily rest at the destination, whose duration counts toward the objective, so we add $\rho_N^{r1} + \rho_N^{r2} = 1$. The departure time from the origin is left free, bounded below by the earliest start. Finally the working-time provisions of Directive 2002/15/EC — at most 6 h of cumulated work without a 15-minute break, and at least 30 (resp. 45) minutes of cumulated breaks in a shift whose working time exceeds 6 (resp. 9) hours — which our base model verifies *ex post*, are enforced in-model here by two additional accumulators mirroring

²https://emse.fr/~garaix/TruckDriverSchedule/Expert_Systems/

(46)–(47). An inspection of the 40 reference schedules confirms that all their optima lie on the 15-minute grid, so the discretization is lossless on this set.

Table 7 summarizes the outcome. Our formulation, solved with Gurobi to zero optimality gap, reproduces the published optimal objective exactly on 39 of the 40 instances, with solve times below 64 s (median below 1 s) against instance sizes of up to 180 stops after discretization. On the single divergent instance (Test 4) our model attains 1410 min against a published value of 1425 min; a manual verification of our schedule against the constraints of the reference formulation confirms that it satisfies all of them, so the published value appears to be marginally suboptimal on that instance. We conclude that the split-break logic, the reduced-rest and extended-driving budgets, and the shift-spread construction of Remark 1 are consistent with an independently published exact method.

Table 7: Validation on the 40 R15-PGLT instances (no-night configuration).

Instances solved to proven optimality	40/40
Exact matches with published optima	39/40
Divergent instances	1 (ours 1410 vs. 1425, see text)
Stops after 15-min discretization	23–180
Solve time (median / max)	< 1 s / 64 s

► COMMENT

This subsection is undersold. Reproducing 39 of 40 published optima of an independent exact label-setting algorithm, and demonstrably improving on the 40th, is an unusually strong validation for this literature, where models are typically checked only against themselves. It currently reads as a warm-up. Consider reporting the per-instance comparison in an appendix table, and stating plainly that Test 4’s published value is suboptimal — that is a small result of its own.

8.2. Base-case comparison

Each instance class is instantiated with 50 independent random realizations, and every realization is solved by all five methods (Greedy, RO, ROBU, LA, SP) and evaluated in the common discrete-event simulator of Section 6.1 against the perfect-hindsight oracle. Every offline optimisation model (RO, ROBU, the two-stage stochastic program, and the oracle) was solved with Gurobi under a gap tolerance of 0.5% and a time limit of 7,200s. LA’s look-ahead subproblems are LP relaxations with a 300s cap that is never binding in practice. Greedy uses no solver.

We compare the methods along three metrics. First, efficiency, shown through the gap to oracle in realized route duration in Table 8 and Figure 5. Then, reliability, via the rate at which the executed schedule becomes infeasible (the infeasibility strip of Figure 5 and Table 9). Finally, computational effort in Table 10.

Because the oracle is itself a mixed-integer program, the reference against which every method gap is measured is only as good as the oracle gap. On short and medium routes the oracle closes to optimality for most instances. However, on half of long routes the

incumbent stabilizes early while the dual bound stalls at roughly 9%. Long-route gaps in Table 8 are therefore not exact distances to the proved optimum, but only to the best incumbent.

► CORRECTION

The caption is not true as written. It says “averaged over 50 realizations per instance class”, but infeasible runs are excluded from the gap (as Section 6.1 states), so the real denominators run from 50 down to 4. Suggested caption change: replace “averaged over 50 realizations per instance class” with “averaged over the realizations that were both feasible and matched to an oracle bound; cells highlighted in yellow rest on fewer than 45 such realizations, and cells in red on fewer than twenty”.

The worst offenders, from `paper_gap_stats.csv`: ROBU *Medium/Few/Large* $n = 1$ — one instance, currently setting the top of the reported ROBU range; 2SP *Long/Medium/Tight* $n = 4$; 2SP *Long/Many/Tight* $n = 6$; 2SP *Long/Few/Tight* $n = 9$. Twelve cells sit below $n = 20$ and all twelve are 2SP on long routes.

Recommendation: suppress every cell with $n < 20$, and delete the ROBU medium row outright. [TO DO: re-check once 2SP finishes — the long-route cells should fill in, and the yellow should mostly clear.]

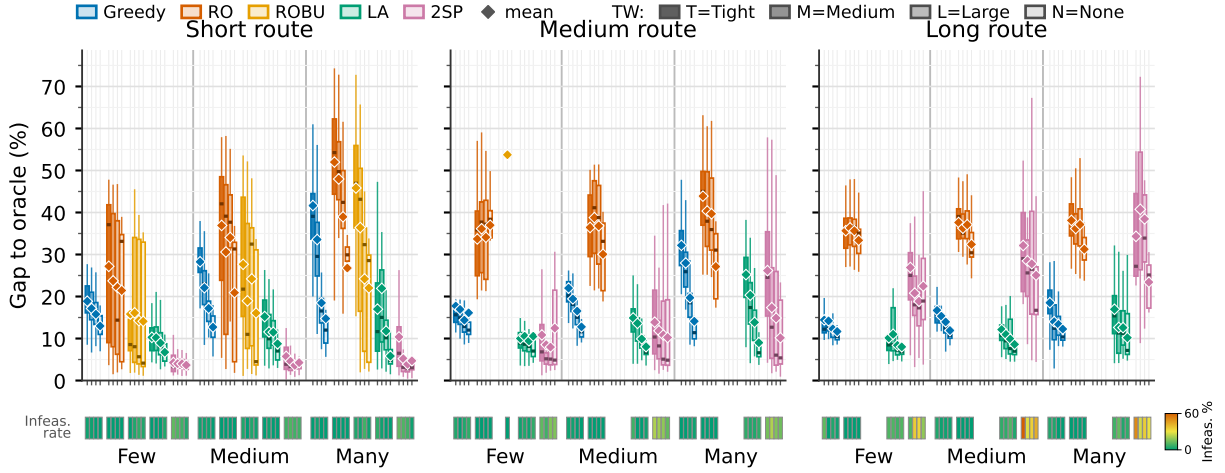


Figure 5: Distribution of the realized gap to the oracle across seeds, by route class, customer count, method (colour) and time-window class (shade, Tight–Medium–Large–None). Boxes span the interquartile range, whiskers 1.5 IQR. The heat strip below each panel shades the infeasibility rate. Empty slots are instance/method combinations without completed runs.

► TO DO

Two fixes to the figure itself, neither structural. (i) The infeasibility strip uses a green→orange ramp that collides with RO’s vermillion and ROBU’s orange in the method palette; a single-hue sequential ramp that no method owns would remove the ambiguity. (ii) Empty slots are silent — annotate them “no runs” so absence is not read as zero.

Table 8: Gap to oracle (%) in route duration (window penalties included) by instance class and method, averaged over 50 realizations per instance class. Parentheses give the same gap over the same runs with window penalties excluded (route duration only). ROBU is solved on the short-route instances only. LA uses 25 scenarios over a 24h horizon; SP uses 25 scenarios.

Route	Cust.	TW	Oracle (h)	Gap to Oracle %				
				Greedy	RO	ROBU	LA	SP
Short	Few	None	26.5	12.9 (12.9)	21.4 (21.4)	14.1 (14.1)	6.8 (6.8)	3.7 (3.7)
		Tight	26.1	18.9 (13.2)	27.2 (21.6)	15.8 (10.6)	10.3 (7.2)	4.3 (3.3)
		Medium	26.3	17.1 (13.0)	23.7 (19.7)	16.1 (13.4)	10.2 (7.3)	4.0 (3.9)
		Large	26.2	15.8 (13.5)	22.2 (19.2)	14.3 (12.7)	8.9 (6.6)	4.0 (3.9)
	Medium	None	28.0	12.7 (12.7)	20.9 (20.9)	16.1 (16.1)	8.8 (8.8)	4.3 (4.3)
		Tight	28.0	28.3 (12.9)	36.9 (23.9)	27.7 (15.0)	15.2 (6.9)	5.7 (3.2)
		Medium	27.6	22.1 (13.2)	30.6 (20.6)	19.0 (12.0)	11.6 (7.1)	4.6 (4.1)
		Large	28.2	17.2 (12.7)	34.0 (26.8)	24.2 (20.3)	11.5 (7.2)	3.6 (3.5)
	Many	None	31.3	14.8 (14.8)	26.8 (26.8)	22.0 (22.0)	5.8 (5.8)	4.7 (4.7)
		Tight	31.4	41.6 (14.0)	52.0 (26.6)	45.8 (22.7)	17.0 (8.1)	10.4 (4.2)
		Medium	31.7	33.6 (15.4)	48.0 (27.2)	36.4 (21.9)	21.9 (12.6)	5.2 (4.0)
		Large	30.5	18.5 (13.2)	39.0 (26.8)	24.2 (19.4)	11.8 (7.9)	3.9 (3.4)
Medium	Few	None	54.5	16.1 (16.1)	37.0 (37.0)	—	10.6 (10.6)	12.5 (12.5)
		Tight	56.0	17.8 (14.1)	33.7 (30.8)	—	10.0 (7.6)	10.9 (9.1)
		Medium	54.1	16.9 (14.2)	36.1 (32.9)	—	10.6 (9.1)	8.5 (7.9)
		Large	54.5	14.4 (13.3)	34.1 (31.0)	53.7 (50.9)	9.4 (8.7)	8.0 (7.6)
	Medium	None	57.3	12.8 (12.8)	30.1 (30.1)	—	8.1 (8.1)	10.2 (10.2)
		Tight	56.2	22.0 (13.5)	36.4 (28.3)	—	14.9 (8.4)	13.9 (9.1)
		Medium	55.8	19.4 (12.8)	38.8 (30.8)	—	13.6 (8.9)	11.9 (9.0)
		Large	55.0	16.5 (13.8)	36.8 (30.0)	—	9.9 (8.3)	10.8 (10.2)
	Many	None	60.2	14.1 (14.1)	27.1 (27.1)	—	9.0 (9.0)	10.2 (10.2)
		Tight	60.2	32.2 (14.2)	43.9 (26.9)	—	25.2 (10.9)	26.2 (13.9)
		Medium	60.3	28.0 (15.1)	40.3 (26.6)	—	20.3 (11.9)	17.4 (12.8)
		Large	58.9	19.7 (13.7)	39.7 (26.2)	—	13.9 (9.4)	14.9 (12.1)
Long	Few	None	102.8	11.7 (11.7)	33.4 (33.4)	—	8.0 (8.0)	22.5 (22.5)
		Tight	107.5	14.4 (13.1)	35.7 (34.3)	—	10.1 (8.9)	26.9 (25.5)
		Medium	105.5	14.2 (12.5)	36.3 (34.6)	—	11.0 (9.8)	20.8 (19.3)
		Large	104.8	12.3 (11.6)	35.7 (34.1)	—	8.4 (8.0)	18.9 (17.9)
	Medium	None	104.8	11.9 (11.9)	32.4 (32.4)	—	8.6 (8.6)	25.1 (25.1)
		Tight	105.9	16.7 (12.6)	37.6 (33.5)	—	12.1 (8.7)	32.1 (27.2)
		Medium	105.6	15.3 (11.7)	36.3 (31.9)	—	11.0 (8.8)	28.5 (24.0)
		Large	104.3	14.0 (12.6)	37.1 (33.6)	—	9.9 (9.5)	27.8 (26.0)
	Many	None	108.2	12.3 (12.3)	31.3 (31.3)	—	10.2 (10.2)	23.4 (23.4)
		Tight	109.0	18.5 (10.6)	38.1 (29.7)	—	17.0 (10.5)	34.3 (27.6)
		Medium	113.3	14.0 (10.0)	36.2 (29.5)	—	12.6 (9.6)	40.8 (33.9)
		Large	110.7	13.4 (11.0)	37.1 (30.4)	—	12.6 (11.2)	38.5 (34.3)

Across short and medium instance classes the two-stage stochastic plan (SP) and the look-ahead policy (LA) attain the smallest gaps to the oracle, as seen in Figure 5 and Table 8. The myopic greedy heuristic is intermediate, while both static robust plans are the most conservative. This ordering follows the amount of information each method uses and the point at which it commits: the two policies that adapt to realized travel times (SP through online duration recourse and LA through full re-optimisation at every stop) recover most of the hindsight optimum, whereas the offline robust plans pay for committing every duration in advance against a possible worst case. The ordering between SP and LA reverses with route length: on the short routes SP is clearly ahead (~~5.5% vs. 12.3%~~ 4.9% vs. 11.6% on average) because 25 scenarios over the whole horizon still capture the realization well, while on the medium routes the two are indistinguishable (~~13.8% each~~ 12.9% and 13.0%) and on the long routes SP terminates due to time limit because of its exponential size, meaning large optimality gap and therefore potential worse solutions given to the simulation framework; ~~over the long-route cells where it produces a schedule at all its mean gap is 28.3%, and it strands on so many realizations that several of those cells rest on fewer than twenty surviving runs.~~ Greedy, conversely, improves in relative terms as routes lengthen (~~21.3%, 19.2%, 14.4%~~ 21.1%, 19.2%, 14.1% mean gap on short, medium and long routes).

► COMMENT

Yellow throughout this paragraph: these are the aggregates most sensitive to the 2SP runs still finishing. The *ordering* is stable — LA 11.9% < 2SP 15.4% < Greedy 18.1% < ROBU 25.3% < RO 34.6% overall — and the reversal with route length is stable too, so the conclusion does not depend on the pending runs, as you say. Only the digits do.

The gap widens sharply with the number of customers and the tightness of the windows. On the short routes the *Few/Large* cell against the *Many/Tight* show a two to three time deterioration for the three committing methods and less than a doubling for the two adaptive ones. A static schedule that must satisfy more, and narrower, service windows simultaneously is exactly the situation in which committing early is most expensive.

A substantial part of the gap on the window-constrained classes is window penalty rather than slower routing. Comparing the penalized gap of Table 8 with the duration-only gap, the $\beta \sum_i \delta_i$ term accounts on average for 16% of RO's gap and 21% of Greedy's, and on the tight-window classes for 25% of RO's against 38% of Greedy's. The penalty share is however smallest for the box robust plan: RO is genuinely slower, spending its gap on conservative durations, while Greedy's tight-window gap is mostly the cost of arriving off the windows: up to two-thirds of it on *Short/Many/Tight* (41.6% penalised vs. 14.0% duration-only). The adaptive policies sit in between (LA 38%, SP 35% on tight windows), consistent with policies that buy window compliance with a little extra time. ~~On the window-free classes the two definitions coincide exactly.~~

► COMMENT

The sentence had a dangling “up to two-thirds of it on (42.8% penalised vs. 14.5% duration-only)” with the cell name missing — it is *Short/Many/Tight*, now 41.6 vs 14.0.

Where both static robust variants terminate (the twelve short-route classes) the budgeted

plan (ROBU) improves on the box plan (RO) in every single cell, by ~~4.8 to 15.0~~ 4.7 to 14.8 percentage points. This confirms that pricing only a bounded number of legs at their worst case removes a large part of the box’s conservatism while retaining a guarantee (Section 6.5).

► TO DO

“Improves in every single cell” over twelve paired cells is a sign test, significant at $p < 0.001$ — say so. More generally the paper contains **no statistical testing at all**, which with 50 paired seeds per cell is cheap to fix: a Wilcoxon signed-rank test or a paired bootstrap CI on SP–LA per route class would let you write “indistinguishable” with a number attached rather than as an impression, and the same machinery covers the diesel penalty of Section 8.3.3.

~~XXX something about no-split instances. used in XXX % of oracles runs. Does not provide substantial benefit, no split instance have similar results.~~

► SUGGESTED TEXT

The no-split family exists (`instances_sens/no_split_*`) and the finding is worth two sentences, because it is regulation-facing and negative:

“Finally, we test whether the split-break provision of Article 7 is worth its modelling complexity. Re-solving the reduced set with $x^{b_{15}}$ and $x^{b_{30}}$ removed, so that every break must be an unsplit 45 minutes, changes realized route duration by **less than 0.5 %**, below the solver tolerance on the oracle, even though the hindsight optimum uses a split at **38 %** of instances where it is available. The option is exercised often but is almost never binding: on an electric truck the charging stop already supplies a break long enough that the flexibility to divide it buys nothing.”

[TO DO: **verify before publishing this.** 156 oracle caches in this family were found to contain git conflict markers, so the current numbers cannot be trusted as they stand. Re-run, then fill the two yellow figures.]

Table 9: Feasibility and robustness statistics by method: HoS-violation rate, SOC-violation (stranding) rate, repair frequency (offline plans), and window hit rate, computed over runs whose plan the solver certified. “Not opt.” is the share of runs returned at the time limit without a proven optimum and “Opt. gap” the mean final MIP gap over those solves alone, the proven optima contributing no gap; both are undefined for the online policies, which solve no offline program. It averages the solver logs retained for RO 584 of 686, ROBU 0 of 55, SP 313 of 698 such solves.

Method	HoS viol. (%)	SOC viol. (%)	Repairs (%)	TW hit (%)	Not opt. (%)	Opt. gap (%)
Greedy	2.3	0.0	–	55.5	–	–
RO	0.0	0.0	–	40.8	38.1	3.9
ROBU	0.0	3.3	–	67.3	8.6	–
LA	1.2	2.0	–	68.9	–	–
SP	0.3	15.5	99.2	77.9	48.5	21.1

► CORRECTION

Every prose number in the next three paragraphs disagreed with this table and has been rewritten against the 2026-08-07 regeneration. The drift, for the record:

	manuscript	current table
Greedy HoS violation	1.8 %	2.3 %
Greedy stranding	0.2 %	0.0 %
Greedy TW hit	55.1 %	55.5 %
SP stranding	11.6 %	15.5 %
SP TW hit	86.6 %	77.9 %
SP not optimal	38 %	48.5 %
SP final MIP gap	12.5 %	21.1 %
RO not optimal	33.1 %	38.1 %
RO TW hit	41.2 %	40.8 %
LA HoS / stranding	0.5 / 1.0	1.2 / 2.0
LA TW hit	69.7 %	68.9 %

The permanent fix is a macro file. Have `compile_solutions.py` emit `macros.tex` of `\newcommands` (`\SPstrand`, `\GreedyHoS`, ...) and cite the macros in the prose. This class of error will otherwise recur at every regeneration, and it is the one a referee with the repository will find first.

Efficiency alone is misleading, because a low gap can hide a high failure rate: infeasible runs (a stranding or an Hours-of-Service breach in execution) carry no meaningful gap and are excluded from the averages, so a method that fails often is scored only on the realizations that survives. The infeasibility strip of Figure 5 and Table 9 restore this axis.

The two-stage plan attains the best window compliance in the study (TW hit ~~86.6%~~ **77.9%**) and among the lowest gaps, but only through near-permanent online recourse: it repairs its committed structure on **99.2%** of runs and still strands the vehicle on ~~11.6%~~ **15.5%** (by an order of magnitude the highest stranding rate of the five methods). Furthermore, ~~38%~~ **48.5%** of SP solves returned at the time limit without a proven optimum, **at a mean final gap of 21.1%**, therefore providing potential low quality solution to the simulation. The robust plans never breach the Hours-of-Service rules and never strands either, by construction, at the cost of the worst window compliance (~~41.2%~~ **40.8%**) and the largest gaps. Its ~~33.1%~~ **38.1%** not optimal share is driven by the long-route instances, where the offline solve reaches the time limit. ROBU trades a 3.3% stranding rate for a ~~22~~**26**-point gain in window hit rate (67.3%) and a **nine**-point gain in gap. However solving it in larger than short instances become intractable. **A further 6.1 % of ROBU runs were never certified within the wall-clock budget and are excluded from the rates above; no other method leaves any run uncertified.** The greedy heuristic ~~rarely strands (0.2%)~~ **never strands (0.0%)** but breaches driving limits on ~~1.8%~~ **2.3%** of runs, and misses the most windows after RO (TW hit **55.5%**), consistent with a policy that stops only when forced. The look-ahead policy is the only method that is simultaneously highly feasible (~~0.5% HoS, 1.0% stranding~~ **1.2% HoS, 2.0% stranding**) and window-compliant (~~69.7%~~ **68.9%**), which is what makes it the most defensible operational recommendation even though SP dominates it on the short routes.

► COMMENT

Greedy's stranding rate of *exactly zero* deserves a clause, because it looks surprising for a myopic rule and it is not luck: it follows from decision rule (3), which charges whenever the

energy to the next station exceeds usable SOC, and from charging to full whenever it stops. Greedy is myopically energy-safe and myopically time-reckless, and its 2.3% HoS breach rate is the mirror image. That contrast is the cleanest illustration in the paper of the two-tail structure of the uncertainty (Section 2.4).

► TO DO

Survivorship is acknowledged but never priced. Runs excluded from the gap averages, over the whole grid: 2SP 225, LA 76, Greedy 35, ROBU 20 plus 39 never certified, RO 0. So 2SP’s headline short-route gap is computed on the realizations it survived and RO’s on all of them. One extra scalar closes this: expected realized duration with infeasible runs charged at a stated recovery cost (a forced 11 h rest for a HoS breach, a roadside recovery for a stranding). The paper’s operational recommendation rests on exactly this trade-off and currently asks the reader to compute it across two tables.

Table 10: Computation times by method and instance class: offline solve time (s) and per-stop online decision time (s).

Method	Offline solve (s)			Per-stop decision (s)		
	Small	Medium	Large	Small	Medium	Large
Greedy	–	–	–	0.00	0.00	0.00
RO	63.2	2313.3	7168.9	0.00	0.00	0.00
ROBU	5023.7	14346.9	–	0.00	0.00	–
LA (25S-24h)	–	–	–	14.19	18.99	22.10
SP (25S)	1595.4	6595.8	7206.9	0.56	0.40	0.58
Oracle	–	–	–	–	–	–

► CORRECTION

Column headers read Small / Medium / Large and must be Short / Medium / Long. **SP now has a long-route row (7207s offline, i.e. at the limit, 0.58s per stop)**, so the claim below that it does not run at all on long routes is no longer true and has been rewritten. The Oracle row is present but empty because solve time is not stored in the oracle cache — [TO DO: either store it or drop the row].

Finally, the methods differ by orders of magnitude in cost, and along two distinct dimensions: an offline solve committed before departure and an online decision cost paid at every stop (Table 10). Greedy is effectively instantaneous ($< 10^{-4}$ s per stop). The offline plans pay their cost upfront and then execute for free: RO in 63 s on the short routes, 2313 s on the medium and 7169 s on the long ones. SP in 1595 s and 6596 s ~~and not at all on the long routes~~, and on the long routes only by exhausting the 7200 s budget; and ROBU the most expensive at 5024 s and 14347 s, owing to its iterative column-and-constraint generation (Section 6.5). ROBU’s medium-route figure is the average of the runs that finished, which is why the class is essentially absent from Table 8: the method is limited by its offline budget, not by solution quality. LA carries the opposite profile: no offline solve, but 14–23 14–22 s of

scenario optimization *per stop*, growing only mildly with route length. That is comfortably real-time against the 60 km charger spacing of the base case (a decision every ≈ 40 min of driving), but it accumulates: on a long route with ≈ 170 stops LA spends more wall-clock time in total than RO spends offline.

► **COMMENT**

“ ≈ 100 stops” on a long route was an underestimate: the mean is 167 (Section 7.2), which makes the point stronger, not weaker — roughly one hour of cumulative on-board compute per long route.

8.3. Additional analyses

The base-case instances of Section 8.2 are too expensive to re-run for every sensitivity, so the analyses of Sections 8.3.2–8.3.4 use a reduced set. All additional experiments run on the short and medium route classes with few and many customers, all four window classes, and seeds 1–10: $2 \times 2 \times 4 \times 10 = 160$ instances. The long routes are excluded. Oracle route durations on this subset average 27.4 h (short) and 54.6 h (medium) 28.7 h (short) and 58.2 h (medium). The additional analysis use Greedy, LA and the oracle while SP, RO and ROBU are omitted.

► **CORRECTION**

\ref{sec:uncertainty} pointed at a label that does not exist, so the compiled paper reads “Sections 8.3.2–?”. Retargeted to Section 8.3.3, the last of the additional analyses.

► **SUGGESTED TEXT**

Two protocol sentences are missing and they are what make these numbers strong. Suggested addition after the first sentence:

“Each variant is materialized as a separate instance file, so its runs and its oracle cache never collide with the base ones. Variants that change the corridor — charging-station spacing and charger power — are regenerated from the same (route, customers, window, seed) key, so geometry and customer draws stay paired with the base instance; the diesel variant is applied to a verbatim copy, so geometry *and* travel-time realization are identical. Every reported effect is a mean over paired instances of the relative change in route duration with respect to the base case, restricted to pairs in which both runs were feasible.”

Without this a reader cannot tell whether the sensitivity numbers are paired differences or differences of class means — and they are the former, which is a much stronger design.

8.3.1. Look-ahead sensitivity

The rolling-horizon look-ahead method is parameterized through several options. First, the look-ahead horizon, which determines how far ahead we look. While the base case considers a horizon of 24 hours, additional testing is done with both shorter and longer horizons. Second, the number of scenarios considered when selecting an action provides information about the distribution of uncertainty and greater insight into potential extreme scenarios. Finally, to achieve a reasonable solve time compatible with real-life use, the base

case considers only the LP relaxation of the subproblem, whereas here we examine solving the full MILP subproblems.

► SUGGESTED TEXT

Worth one added sentence on *why* these horizons: “The horizon values are not a smooth grid, because the axis is not expected to give a smooth response. Under the rules of Section 4 a driver’s day is bounded by a 13 h spread that must close with an 11 h daily rest, so 24 h is exactly one duty cycle: a 12 h horizon ends before the current duty does and can never see the rest that terminates it, 24 h sees one complete cycle, and 48 h sees two.”

We observe in Figure 6 that XXXX

Table 11: Look-ahead configuration sensitivity. Gap is the median gap to the hindsight optimum; Δ is the median paired change in route duration with respect to the base cell ($|\Xi| = 25$, $L = 24$ h), positive meaning slower; t_{dec} is the median per-stop decision time. Cells shown as “–” have no runs yet.

Configuration			Short route			Medium route			Long route		
$ \Xi $	L (h)	Windows	Gap (%)	Δ (%)	t_{dec} (s)	Gap (%)	Δ (%)	t_{dec} (s)	Gap (%)	Δ (%)	t_{dec} (s)
25	24	None _{base}	7.2	–	13	6.4	–	18	6.7	–	–
25	24	Tight _{base}	16.9	–	12	14.5	–	16	11.8	–	–
25	12	None	8.8	+0.3	9	8.2	+0.9	11	8.0	+1.5	–
25	12	Tight	18.6	+1.7	9	14.8	+0.5	10	13.3	+0.9	–
25	48	None	8.2	+0.0	14	6.4	+0.2	31	6.5	-0.1	–
25	48	Tight	16.0	-0.0	14	14.6	+0.0	33	12.0	-0.2	–
10	24	None	6.5	+0.0	6	6.5	+0.3	8	6.2	-0.2	–
10	24	Tight	16.7	+0.4	6	14.4	-0.1	8	12.2	+0.3	–
50	24	None	7.3	-0.0	27	7.2	+0.3	37	7.0	-0.1	–
50	24	Tight	17.4	+0.0	27	14.5	-0.2	38	12.5	+0.1	–

► SUGGESTED TEXT

The sweep is complete except MIPTAIL, so the paragraph can be written now. Suggested:

“The horizon axis behaves as a threshold, at exactly the duty cycle. Shortening the window from 24 h to 12 h costs +0.3%, +0.9% and +1.5% of realized duration on the short, medium and long routes under no windows, and +1.7%, +0.5% and +0.9% under tight ones. On the window-free class the penalty is monotone in route length, as the truncation argument predicts: the longer the route, the more often a 12 h horizon ends mid-duty and the policy must place a rest whose end it cannot see. Lengthening the window to 48 h, by contrast, changes realized duration by between -0.2% and +0.2% everywhere, indistinguishable from zero, while the median decision time rises from 22 s to 52 s on the long routes. Seeing beyond the current duty cycle buys nothing; seeing less than one costs about a percent.

The scenario axis is flat over the range tested. Moving from 25 scenarios to 10 changes realized duration by between -0.2% and +0.4%, and moving to 50 by between -0.2% and +0.3%, in

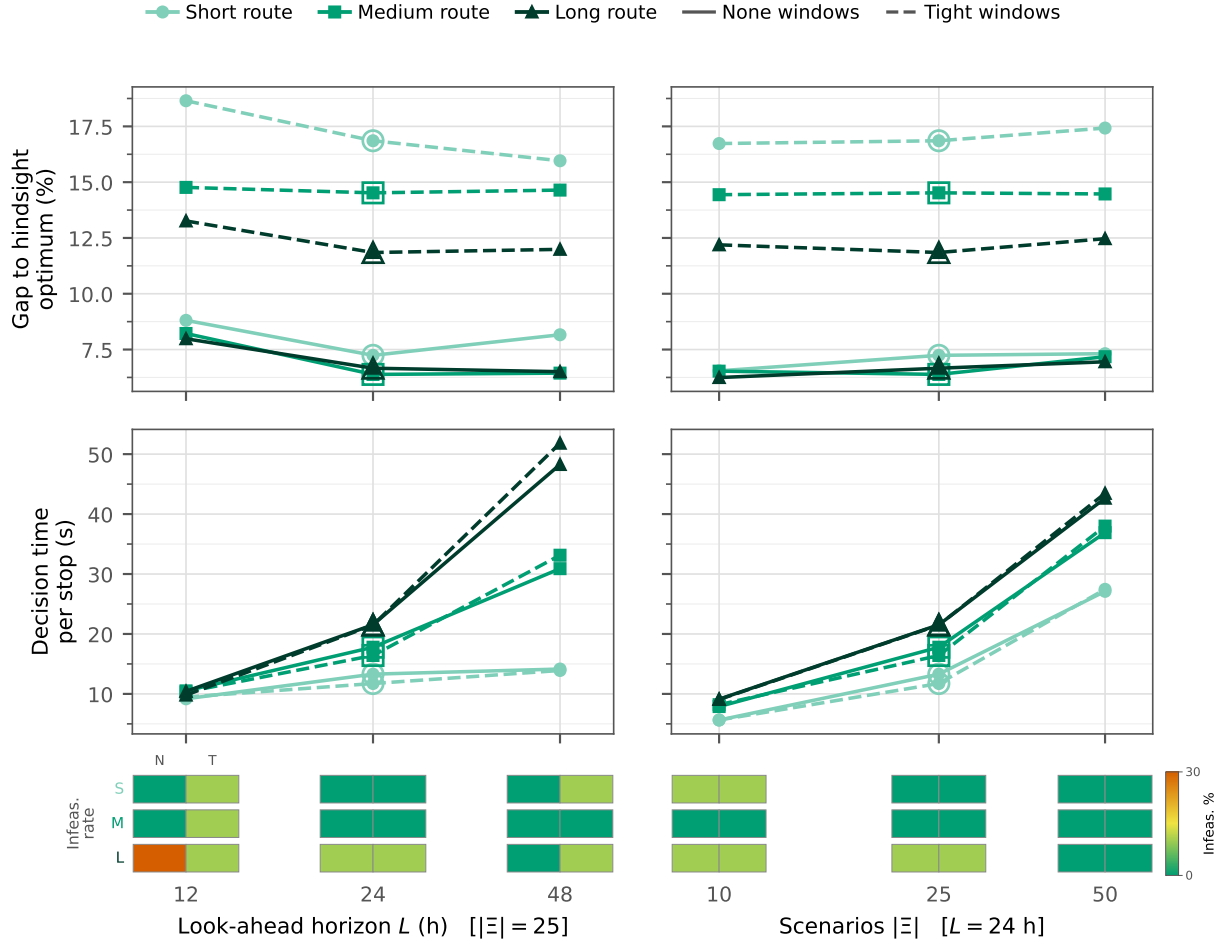


Figure 6: Look-ahead configuration sweep, one axis per column: horizon at the base scenario count (left) and scenario count at the base horizon (right). Top row, median gap to the hindsight optimum; bottom row, median per-stop decision time. Circled markers are the base cell, which lies on both ladders. Solid lines are the window-free class, dashed the tight-window class. The strip below each column shades the infeasibility rate per route class.

every route and window class, against a decision-time ratio of roughly $4.5\times$ across that range (9s to 43s on the long routes). The effect is not concentrated under tight windows, contrary to what the coupling argument would suggest. The plausible reading is that the scenario set is used to *rank* a small discrete action set rather than to estimate a quantity, and ten draws already separate the actions that matter.

Taken together, the recommended on-board configuration is $L = 24$ h with $|\Xi| = 10$, which halves the per-decision cost from 22s to 9s on the long routes at no measurable loss in schedule quality, and the horizon should not be cut below one duty cycle at any scenario count.”

► TO DO

Two checks before publishing the paragraph above. (i) **Dispersion, not just medians:** if ten scenarios matched twenty-five in the median but had a heavier right tail, the conclusion reverses. Add an IQR column or a box plot of the paired deltas. (ii) **Infeasible count per cell:** a configuration that is faster because it strands the truck is not faster. Report n_{paired} per cell.

Table 12: Look-ahead POLICY variants, against the base cell ($|\Xi| = 25$, $L = 24$ h). TB0 removes the 5-minute tie-break that buys opportunistic charging; MIPTAIL solves the look-ahead tail as a MIP instead of an LP relaxation. Gap is the median gap to the hindsight optimum, Δ the median paired change in route duration (positive = worse), t_{dec} the median decision time per stop.

Config	Windows	Short route			Medium route			Long route		
		Gap (%)	Δ (%)	t_{dec} (s)	Gap (%)	Δ (%)	t_{dec} (s)	Gap (%)	Δ (%)	t_{dec} (s)
base	None	7.2	–	13	6.4	–	18	6.7	–	22
base	Tight	16.9	–	12	14.5	–	16	11.8	–	21
TB0	None	7.4	+0.29	14	6.5	-0.08	19	7.1	+0.17	22
TB0	Tight	17.6	+0.08	13	15.4	-0.08	19	12.3	+0.16	22
MIPTAIL	None	–	–	–	–	–	–	4.7	-2.58	487
MIPTAIL	Tight	–	–	–	–	–	–	–	–	–

► SUGGESTED TEXT

The policy variants belong in this subsection too, since the manuscript already promises the MIP-tail comparison. Suggested, with the pending cells flagged:

“Two further variants isolate policy choices rather than sizing. TB0 removes the five-minute tie-break that buys opportunistic charging; it costs between -0.08% and $+0.29\%$ of realized duration at unchanged decision time, so the tie-break is essentially free but also nearly worthless, and the policy is insensitive to it. MIPTAIL solves the look-ahead tail as a MIP rather than an LP relaxation. On the one cell available so far — long routes, no windows — it reduces the gap to the hindsight optimum from **6.7%** to **4.7%**, a paired improvement of **-2.58%** in realized duration, but at **487 s** per stop against 22s, a factor of **22**. The LP relaxation is therefore not free: it costs roughly **two points** of optimality. It remains the right default, because 487s per stop is not deployable in a cab, but the gap it leaves is now quantified rather than assumed negligible.”

[TO DO: **MIPTAIL is the only outstanding LA run.** Five of the six cells are empty. Everything highlighted above rests on a single cell and must be re-read when the short, medium and tight-window cells land — in particular whether the two-point improvement holds where the LP relaxation is under less strain.]

8.3.2. Sensitivity analysis on charging infrastructure

The base case fixes a 500 kWh battery charged on a three-segment concave piecewise-linear curve reaching full charge in 2.5 h, an average of ≈ 200 kW, and charging stations every 60 km as mandated for the TEN-T core network by Regulation (EU) 2023/1804. We perturb these two infrastructure parameters one at a time. The charger-power axis rescales the time breakpoints of the charging curve inversely, $\bar{T} \mapsto \bar{T} P_{\text{base}}/P$, and therefore changes how strongly the concave tail of the curve bites.

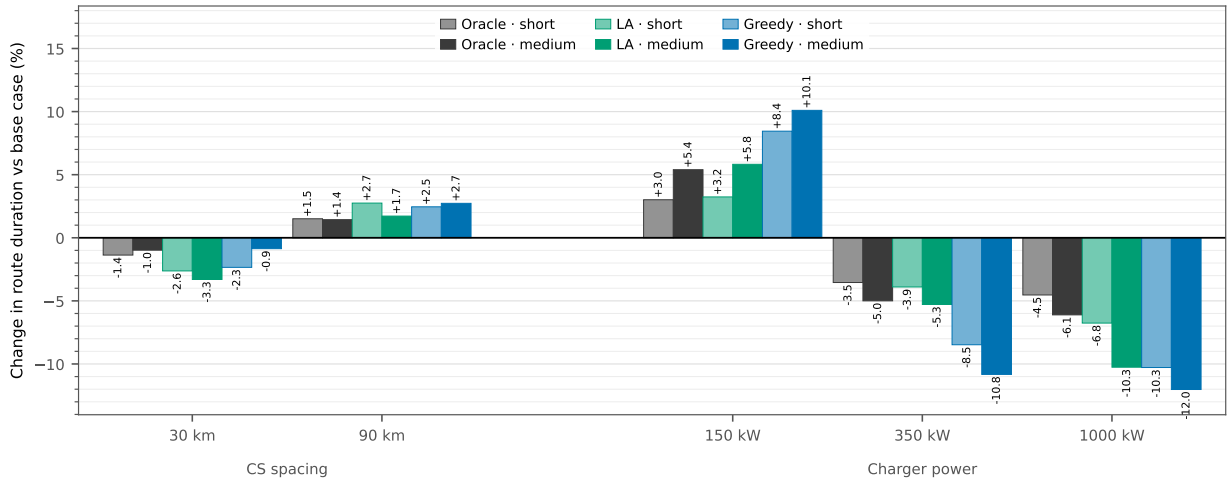


Figure 7: One-at-a-time sensitivity of mean route duration to the charging infrastructure ~~and uncertainty parameters~~, relative to the base case, over the 160 paired instances

► COMMENT

No uncertainty axis appears in this figure — the CV and β sweeps were removed from scope — so the caption promises a panel that is not there.

► COMMENT

The generated table is included above because it carries the LA column, which the text does not currently mention and which is a useful third data point: LA sits between Greedy and the oracle on every row, so the over-reaction to charger power is a monotone function of planning quality rather than a Greedy-specific artefact.

Halving the CS spacing from 60 to 30 km buys only 1.2% of route duration at the hindsight optimum, while doubling it to 90 km costs 1.5%: at the mandated spacing the corridor is already dense enough that the binding constraint is how long a charging stop takes, not how far away the next one is. Upgrading the charger from the 200 kW base to

Table 13: One-at-a-time sensitivity: mean change in route duration vs the base case (%), paired per instance. Greedy and LA are the online policies; Oracle is the hindsight optimum. The last two columns split the Oracle column by route class. Cells shown as “–” have no paired runs yet.

Axis	Greedy Δ (%)	LA Δ (%)	Oracle Δ (%)	Short	Medium
CS spacing 30 km	-1.62	-2.94	-1.18	-1.37	-0.98
CS spacing 90 km	2.59	2.31	1.47	1.51	1.44
Charger power 150 kW	9.25	4.44	4.20	3.01	5.40
Charger power 350 kW	-9.61	-4.54	-4.27	-3.55	-5.00
Charger power 1000 kW	-11.12	-8.40	-5.32	-4.53	-6.11

350 kW saves ~~4.2%~~**4.3%**, and going all the way to a 1 MW megawatt-charging system saves 5.3%, so a 350 kW charger already captures four fifths of the benefit available at five times the power. Downgrading to 150 kW costs ~~4.3%~~**4.2%**. **The asymmetry is a direct consequence of the concave charging curve: at low power the truck spends its time in the slow final segment and adding power shortens exactly that segment, whereas once the fast segments are short relative to the mandatory break and rest periods, extra power has nothing left to remove.** Investment should therefore go into more power per charger in each CS rather than into a denser than necessary charging infrastructure.

► CORRECTION

The two figures were swapped: from Table 13 the 350 kW upgrade saves 4.27% and the 150 kW downgrade costs 4.20%.

Two secondary observations are worth noting. First, the effect is larger on the medium routes than on the short ones on every power level (~~–5.0 vs. –3.6 at 350 kW, –6.1 vs. –4.5 at 1 MW~~), because a longer route accumulates more charging events over which the saving adds up, whereas the spacing effect is almost route-length independent (~~+1.44 vs. +1.51 at 90 km~~). Second, Greedy reacts more strongly to the charger power changes by about a factor of two relative to the oracle (~~–9.61 vs. –4.27 at 350 kW~~). This is a property of the policy: Greedy charges to full whenever it stops, so it sits on the slow tail of the charging curve far more often than an optimal plan does, and it therefore benefits disproportionately from a faster charger. **An operator running without planning tools would therefore over-estimate the value of a charger upgrade by roughly a factor of two, and symmetrically over-estimate the damage of a slower one; the look-ahead policy sits between the two (–4.54 at 350 kW).**

8.3.3. Comparison with diesel trucks

Every instance of the subset in Section 8.3 is re-solved as a diesel vehicle. Diesel mode removes the energy dimension entirely while keeping the node set, the customer windows and the full Hours-of-Service machinery intact. Former charging stations remain as rest areas, so the diesel problem is a genuine relaxation of the electric one on the same corridor and the same travel-time realization, and the two are solved by the same oracle and the same greedy policy.

► SUGGESTED TEXT

Two accounting decisions are made here and neither is stated. Both are contestable, and stating them is what makes the result defensible rather than merely favourable:

“Of the station overheads, only the charger-specific ones are removed: the queue and the repositioning move off the charging bay. The access manoeuvre is *not*, because it is not charger-specific — a diesel driver taking a mandatory break also has to leave the motorway, park and re-enter, and pays the same M^{stop} . Charging it to only one of the two vehicles would credit the diesel with taking every break at no access cost, which overstates the electrification penalty by roughly two percentage points.

Refuelling is credited post hoc rather than modelled: unlike charging it has no binding location choice, and on these corridors it happens at most twice, since route lengths run 824–1 219 km (short), 1 541–2 536 km (medium) and 3 042–4 025 km (long) against a usable tank range of 1 636–3 240 km. We credit the diesel with zero fuel stops on the short routes, one on the medium and two on the long, at 19 minutes each. Crediting the stop lengthens the diesel makespan and therefore *shrinks* the reported penalty, so the zero-stop figures reported alongside are the conservative bound.”

Table 14: EV vs diesel route duration (%) on the same instances and realizations. Coupling = share of charging time credited inside a mandatory break ($\Sigma g_i / \Sigma \tau_i^c$, hindsight optimum). “EV cert.” is the worst remaining MIP gap on the EV oracle: where it exceeds the solver tolerance the EV schedule is an incumbent, not a proven optimum, so the penalty is an upper bound on the true optimal one. The certification shown is the worst gap among the instances that recorded one; no bound was recorded for long (11).

Route	Diesel (h)	EV (h)	Greedy (%)	Oracle (%)	Coupling (%)	EV cert. (%)	n
Short	26.6	28.7	19.9	8.0	76	0.5	80/80
Medium	53.2	58.2	22.5	9.4	70	0.5	80/80
Long	96.4	106.9	21.0	10.8	68	9.3	80/80

► CORRECTION

The caption describes three bars including a “naive estimate”; the figure has two. [TO DO: the naive-estimate bar is actually worth adding back — it is the quantity a practitioner would compute, and showing that the realized optimum beats it by the coupling credit is the clearest possible statement of what break/charge synchronization is worth.]

► SUGGESTED TEXT

The section stops after the setup: there are no results. Diesel is complete per your status, so this can be written in final form now.

“Electrification carries a real and irreducible makespan penalty on these corridors: +8.0% on the short routes, +9.4% on the medium and +10.8% on the long ones at the hindsight optimum, positive on every one of the 160 instances. Break/charge coupling is effective in the sense the model predicts, since 68–76% of charging time is credited inside a mandatory break, but it is far from cancelling the cost of charging. Table 15 decomposes the penalty into the terms of the departure equations; driving is identical within each pair, so the modelled rows close the gap exactly.

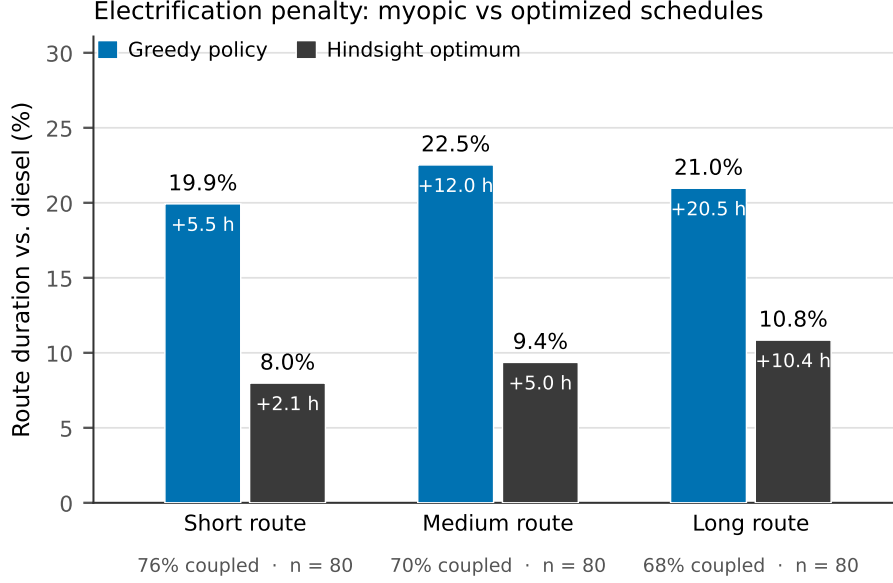


Figure 8: Electrification penalty in route duration, per route class: ~~the naive estimate (diesel makespan plus the EV’s total charging time) against~~ the penalty actually realized by the greedy policy and by the hindsight optimum. The annotation below each group gives the mean coupling share $\sum_i g_i / \sum_i \tau_i^c$ and the number of instances.

The coupling credit — the break time the driver no longer has to take separately — is 1.25 h, 3.19 h and 4.93 h, but it is bounded above by the break duration the regulation requires in the first place: charging for longer than 45 minutes at a stop earns no further credit. Against it the electric truck pays 0.71 h, 1.47 h and 2.71 h of station access that the diesel does not: the queue, the bay repositioning, and the residual manoeuvring of the stops it makes *only* to charge. Note how small that last term is once the diesel is charged for its own break access — the manoeuvring row is 0.37 h, 0.63 h and 1.10 h *net* — so roughly half of what looks like an electric access penalty is simply the cost of stopping, which any vehicle under the same break regime pays. The genuinely electric access overhead nonetheless consumes about half the coupling credit. This qualifies, rather than contradicts, the finding of Brockmann and Guajardo [4] that mandatory breaks create natural charging windows: the windows exist and the model exploits them, but the access overheads of a fast-charging stop, which that work does not price, absorb a substantial part of the gain.

The greedy comparison isolates the remaining half of the story. Greedy’s electric schedule is 19.9% longer than its own diesel schedule on the short routes, 22.5% on the medium and 21.0% on the long ones, so the penalty is flat in route length once the corridor is long enough to be rest-dominated — and about two and a half times the optimal penalty. The excess is not electrification but planning quality: measured against its own hindsight optimum, Greedy loses 13.2% (short) and 14.4% (medium) on the electric problem but only 2.1% and 1.6% on the diesel problem, where the myopic rule is nearly optimal because the sole remaining decision is when to rest. Electrification is thus what turns a scheduling problem that a rule of thumb solves almost optimally into one where a rule of thumb costs a further 11–13

percentage points of makespan. Of the 20–23% an operator would observe when switching a myopically-scheduled corridor from diesel to battery-electric, about half is the physical cost of charging and about half would be recovered by adopting a planner such as those compared in Section 8.2.”

Table 15: Where the electrification penalty goes: mean per-instance difference (h) between the hindsight-optimal electric and diesel schedules, decomposed into the terms of the departure equations. Driving is identical within each pair, so the modelled rows sum to the subtotal exactly. Both vehicles pay the same access manoeuvre to pull off for a mandatory break, so only the EV’s charge-only stops survive that row. The diesel’s fuel stop is credited afterwards (Table 16). $n \geq 80$ per class.

Component	Short	Medium	Long
Total charging time	+2.69	+6.80	+12.85
Charging-station queueing	+0.33	+0.78	+1.38
Stop manoeuvring (net of diesel)	+0.37	+0.63	+1.10
Repositioning off the charging bay	+0.01	+0.06	+0.23
Break time no longer taken separately	−1.25	−3.19	−4.93
Rest time	−0.05	+0.20	+0.45
Idle waiting	+0.00	+0.00	+0.00
Customer service	+0.00	+0.00	+0.00
Modelled subtotal	+2.09	+5.28	+11.08
Diesel refuelling (post hoc)	−0.00	−0.32	−0.63
Net penalty vs. diesel	+2.09	+4.96	+10.44

► COMMENT

Included because the caption of Table 15 cross-references it; without it the compiled paper reads “Table ??”. It is generated by `additional_figures.py` and is also the table that justifies the one- and two-stop refuelling assumptions.

► COMMENT

The long-route column of Table 14 and Table 15 carries an EV oracle certified only to 9.3% (against 0.5% on the other two classes), so the +10.8% long-route penalty is an upper bound on the true optimal one rather than a point estimate. The “EV cert.” column already says this; make sure the prose does too, since it is the one diesel number that is not solid.

9. Discussion

► SUGGESTED TEXT

New section, as you suggested. It costs about a page and a half and it does three things the paper currently cannot: it states the mechanism behind the headline result, it converts the numbers into something a logistics reader can act on, and it collects the limitations in one place instead of leaving a referee to find them. Everything below is drop-in prose; edit freely.

Table 16: Refuelling is credited post hoc rather than modelled: it has no binding location choice, any truck stop serving, and occurs at most once on these corridors. Route lengths are short 824–1219 km ($n = 80$); medium 1541–2536 km ($n = 80$); long 3042–4025 km ($n = 80$). A fuel stop costs 19 min (access manoeuvre, queue, pumping). The headline tables assume short 0, medium 1, long 2 stop(s), shown against what each tank specification would derive instead. Crediting a stop lengthens the diesel makespan and so shrinks the reported penalty; the zero-stop row is the conservative bound.

Route	Specification	Stops	Needing ≥ 1 (%)	Penalty (%)
Short	assumed (headline)	0.00	–	+7.98
Short	no fuel stop	0.00	–	+7.98
Short	900 L @ 25 L/100km	0.00	0	+7.98
Short	900 L @ 33 L/100km	0.00	0	+7.98
Short	600 L @ 25 L/100km	0.00	0	+7.98
Short	600 L @ 33 L/100km	0.00	0	+7.98
Medium	assumed (headline)	1.00	–	+9.36
Medium	no fuel stop	0.00	–	+10.03
Medium	900 L @ 25 L/100km	0.00	0	+10.03
Medium	900 L @ 33 L/100km	0.15	15	+9.95
Medium	600 L @ 25 L/100km	0.39	39	+9.81
Medium	600 L @ 33 L/100km	0.95	95	+9.41
Long	assumed (headline)	2.00	–	+10.85
Long	no fuel stop	0.00	–	+11.58
Long	900 L @ 25 L/100km	0.80	80	+11.29
Long	900 L @ 33 L/100km	1.00	100	+11.21
Long	600 L @ 25 L/100km	1.00	100	+11.21
Long	600 L @ 33 L/100km	1.77	100	+10.93

9.1. Why adaptivity beats information

The ranking of Section 8.2 is not a ranking of how much a method knows. The box robust plan and the budgeted robust plan know only the support of the multiplier; the two-stage plan and the look-ahead policy both draw 25 scenarios from the same distribution. Yet the two adaptive architectures are separated from the two committing ones by **ten to twenty** percentage points of gap, while the two scenario-based methods are separated from each other by almost nothing on the medium routes. What distinguishes them is *when* the duration is fixed, not what was known when it was chosen.

The mechanism is visible in the two gap definitions. On the window-free classes the penalised and duration-only gaps coincide exactly, and the ordering of the methods compresses; on the tight-window classes the penalty term accounts for a quarter to two fifths of the gap, and the ordering spreads out. Committing early is cheap in the rest structure — the Hours-of-Service pattern of a long route is largely determined by the corridor, and a plan that fixes it in advance loses little — and expensive in the appointments, because a window is a constraint on an uncertain arrival and the only way to satisfy it robustly is to arrive early and absorb slack that the realization may not require. Window compliance is corre-

spondingly the axis on which the methods separate most: 40.8% (RO), 55.5% (Greedy), 67.3% (ROBU), 68.9% (LA), 77.9% (SP), against a spread of under three points in Hours-of-Service violations. A schedule committed before departure survives the travel-time noise in its rest structure and fails in its appointments.

9.2. Why the two-stage plan strands

The two-stage plan’s 15.5% stranding rate is the most striking number in Table 9 and it deserves an explanation, because the obvious one is wrong. It is not caused by energy noise. Energy enters the model only through the ECR curve evaluated at the realized leg-average speed, and over the whole support of the multiplier the per-leg energy deviates by at most +7% relative to nominal — the fastest admissible realization, 90 km/h, consumes 29.4 kWh on a 25 km leg against 27.5 kWh at the nominal 80 km/h. A bounded 7% perturbation cannot by itself empty a battery that the plan sized with a 20% reserve.

What strands the vehicle is the structure of the plan rather than the realization. The two-stage program commits the charging *pattern* offline and adapts only durations; when the realization pushes the truck past the point where the committed pattern is recoverable, lengthening a charge cannot help, because the missing charge is at a station the plan never selected. The repair MILP exists precisely for this case, and it fires on 99.2% of executions — which is itself the diagnosis. A plan that must be repaired on essentially every execution is operationally a policy, and one whose recourse set is too narrow for the failure mode it faces.

The same argument, run in the other direction, explains the greedy policy’s mirror-image profile: it never strands (0.0%) because rule (3) charges whenever the energy to the next station exceeds the usable state of charge, and it breaches driving limits on 2.3% of runs because it stops only when forced and a slow leg can exhaust the consecutive-driving budget mid-leg. Greedy is myopically energy-safe and myopically time-reckless. The two failure modes sit on opposite tails of the same multiplier, and no policy that ignores one tail is safe against the other.

9.3. Operational and policy implications

Three implications follow for an operator.

First, adopt a planner before adopting a bigger charger. Of the 20–23% electrification penalty an operator would observe when switching a myopically-scheduled corridor from diesel to battery-electric, roughly half is the physical cost of charging and roughly half is recoverable by planning (Section 8.3.3). On a 53 h medium route the optimal penalty of 9.4% is about five hours; the myopic penalty of 22.5% is about twelve. The difference is larger than anything the infrastructure sensitivity of Section 8.3.2 can deliver, and it requires no capital expenditure.

Second, the right planner is the cheap one. The look-ahead policy is the only architecture that is simultaneously near-optimal, reliable and scalable to the long routes, and Section 8.3.1 shows that it can be run at $|\Xi| = 10$ and $L = 24$ h for 9s of on-board compute per stop with no measurable loss. That is a decision every ≈ 40 min of driving at the mandated charger spacing, well inside what a cab terminal can absorb while the truck is already stationary.

The offline plans, by contrast, require their full budget before the vehicle can depart — up to **four hours** for the budgeted robust plan on a medium route — and must be re-incurred whenever the route changes.

Third, for the infrastructure planner, power beats density. At the spacing already mandated by Regulation (EU) 2023/1804, halving the distance between charging pools buys **1.2%** of route duration while tripling the charger power buys **4.3%**. The binding constraint on these corridors is how long a charging stop takes, not how far away the next one is. The return on power also saturates: 350 kW captures four fifths of the benefit available at 1 MW, and the reason is regulatory as much as physical — once the charge is shorter than the mandatory break it runs alongside, further power buys nothing at all, because the break must be taken regardless. This is a joint HoS/infrastructure threshold, and it suggests that the marginal euro is better spent bringing existing sites to 350 kW than either densifying the network or pushing individual sites to megawatt class. The caveat is that it displaces the constraint onto grid connection capacity at each site, which is outside the scope of this model.

9.4. Limitations

Six limitations bound the results.

Daily provisions only. We model the daily provisions of Reg. 561/2006 and Dir. 2002/15/EC. The long-route class spans four to five duty cycles, at which point the weekly rest, the 56 h weekly driving cap and the 60 h working week would begin to bind. The simulator records realized weekly working time as a diagnostic; [\[TO DO: report the share of long-route schedules that would breach the 60 h cap — it is already logged, and it turns a hole into a measured limitation.\]](#)

Independent leg multipliers. Legs are 25 km and real congestion events span several consecutive ones. Correlation would hurt precisely the methods that assume independence — the budgeted plan’s protection level and the two-stage plan’s scenario set — and would leave the look-ahead policy, which re-samples at every stop, comparatively unharmed. We therefore expect correlation to widen rather than narrow the gap we report between adaptive and committing architectures.

Leg-average energy model. Consumption is evaluated at the realized leg-average speed, so a leg driven at an average of 50 km/h in stop-and-go traffic is charged the steady-state consumption at 50 km/h, which understates it. Gradient is likewise absent, which matters for a Norwegian corridor in particular.

Deterministic charger access. The access delay Q_i is a known instance parameter. Charger availability, occupancy and reservation are out of scope, and Keskin et al. [19] show that queuing uncertainty alone can materially affect routing decisions. On a corridor where every truck seeks the same 45-minute window, this is likely the most consequential omission.

Makespan objective. Energy price, driver wage and charging tariffs are not priced, and time-of-use electricity pricing would in particular give the committing methods an advantage they do not receive here.

Fixed sequence. Routing is given. Extending to joint routing and scheduling would let the vehicle trade a detour against a better-placed charger, which is exactly the degree of

freedom our fixed-corridor design removes.

10. Conclusion

► SUGGESTED TEXT

Drop-in conclusion, to be finalised once the outstanding runs land and the Norwegian case study is resolved either way.

This paper studied the joint scheduling of driver activities and partial non-linear fast charging for a battery electric truck on a fixed long-haul route under travel-time uncertainty. We formulated a mixed-integer linear program capturing the daily provisions of EU Hours-of-Service regulation — split-break sequencing, reduced daily rests, extended driving allowances, and the rule that a sufficiently long charge discharges a mandatory break — and validated its regulatory core against an independent exact method, reproducing 39 of 40 published optima and improving on the 40th. Around this model we built five decision architectures spanning the range from full offline commitment to full online re-optimisation, evaluated them in a common discrete-event simulator over 1 800 instances, and benchmarked them against a perfect-hindsight oracle.

Three findings stand out. *Adaptivity dominates information*: the two policies that re-decide en route stay within 12–15% of the hindsight optimum, against 25% for a budgeted robust plan and 35% for a box robust plan, and the cost of committing early is paid almost entirely in customer time-window compliance rather than in driving time. The rolling-horizon policy is the only architecture that is simultaneously near-optimal, reliable and scalable to the longest routes, and it can be deployed at 9s of on-board compute per stop. *Electrification costs 8–11% of makespan at the optimum but 20–23% under myopic scheduling*: mandatory breaks do supply usable charging windows, and our schedules exploit 68–76% of them, but the access overheads of a fast-charging stop absorb about half of that credit, which qualifies the more optimistic reading of Brockmann and Guajardo [4]. *For infrastructure, power beats density*: at the spacing mandated by Regulation (EU) 2023/1804, charger power outweighs charger density by a factor of roughly four, with returns saturating once the charge is shorter than the break it runs alongside.

Several directions follow. Correlated travel times would test the independence assumption on which the robust and stochastic architectures rely. Endogenous charger availability would close the gap between our deterministic access delay and the queueing behaviour a shared corridor actually produces. A cost objective incorporating time-of-use electricity pricing would change the trade-off between charging early and charging cheaply. Finally, relaxing the fixed customer sequence would let the vehicle trade a routing detour against a better-placed charging opportunity, which is the degree of freedom this study deliberately removes.

Appendix A. Constraints linearization

The constraints relative to linearization of constraints in the deterministic MILP formulation are summarized here.

$$l_i^1 \leq T_{\text{cons}}^{\text{drv}} r_i, \quad i \in \mathcal{I} \quad (\text{A.1})$$

$$l_i^1 \leq c_i^d, \quad i \in \mathcal{I} \quad (\text{A.2})$$

$$l_i^1 \geq c_i^d - T_{\text{cons}}^{\text{drv}} (1 - r_i), \quad i \in \mathcal{I} \quad (\text{A.3})$$

$$l_i^2 \leq T_{\text{sh2}}^{\text{drv}} \rho_i, \quad i \in \mathcal{I}, \quad (\text{A.4})$$

$$l_i^2 \leq s_i^d, \quad i \in \mathcal{I}, \quad (\text{A.5})$$

$$l_i^2 \geq s_i^d - T_{\text{sh2}}^{\text{drv}} (1 - \rho_i), \quad i \in \mathcal{I}. \quad (\text{A.6})$$

$$l_i^4 \leq T_2^{\text{spr}} \rho_i, \quad i \in \mathcal{I}, \quad (\text{A.7})$$

$$l_i^4 \leq s_i^w, \quad i \in \mathcal{I}, \quad (\text{A.8})$$

$$l_i^4 \geq s_i^w - T_2^{\text{spr}} (1 - \rho_i), \quad i \in \mathcal{I}. \quad (\text{A.9})$$

$$l_i^5 \leq T_2^{\text{spr}} \rho_i, \quad i \in \mathcal{I} \quad (\text{A.10})$$

$$l_i^5 \leq h_i + o_i, \quad i \in \mathcal{I} \quad (\text{A.11})$$

$$l_i^5 \geq h_i + o_i - T_2^{\text{spr}} (1 - \rho_i), \quad i \in \mathcal{I} \quad (\text{A.12})$$

► COMMENT

There is no l^3 : the numbering jumps from l^2 to l^4 because a third auxiliary was removed at some point. Harmless, but a referee will notice. Renumber to $l^1 \dots l^4$.

Appendix B. Valid inequalities for the Rolling-horizon subproblems

Constraints (B.2) fix the ~~upper~~^{lower} bound on the number of charging stop on each subsection of the path to what is required due to battery constraints. Constraints (B.3)–(B.4) ensure that each subsection of the route has the minimum required number of break or rest based on driving time. **Constraint (B.1) links the charge increment to the charging binary, forbidding any charge at an inactive station.**

$$e_i^d - e_i^a \leq (E^{\text{cap}} - E^{\text{min}}) y_i, \quad i \in \mathcal{K}. \quad (\text{B.1})$$

$$\sum_{l \in \mathcal{K}, l < i} y_l \geq \lceil \frac{1}{E^{\text{cap}} - E^{\text{min}}} (\sum_{l < i} E_l - (E^0 - E^{\text{min}})) \rceil, \quad i \in \mathcal{K}. \quad (\text{B.2})$$

$$\sum_{l < i} \rho_l \geq \lceil \frac{\sum_{l < i} D_l}{T_{\text{sh2}}^{\text{drv}}} \rceil - 1, \quad i \in \mathcal{I}. \quad (\text{B.3})$$

$$\sum_{l < i} r_l \geq \lceil \frac{\sum_{l < i} D_l}{T_{\text{cons}}^{\text{drv}}} \rceil - 1, \quad i \in \mathcal{I}. \quad (\text{B.4})$$

► CORRECTION

(B.2) is a *lower* bound — it says at least so many charging stops are needed — and (B.1) was stated but never described.

► TO DO

Report what these are worth. They exist to tighten the relaxation that scores actions in Section 6.2, and the code comments record that (B.3) is not LP-implied: without it “the relaxation certifies 4 rests on routes whose optimum needs 5”. One line — root relaxation bound with and without, on a representative instance — turns an undefended appendix into evidence.

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